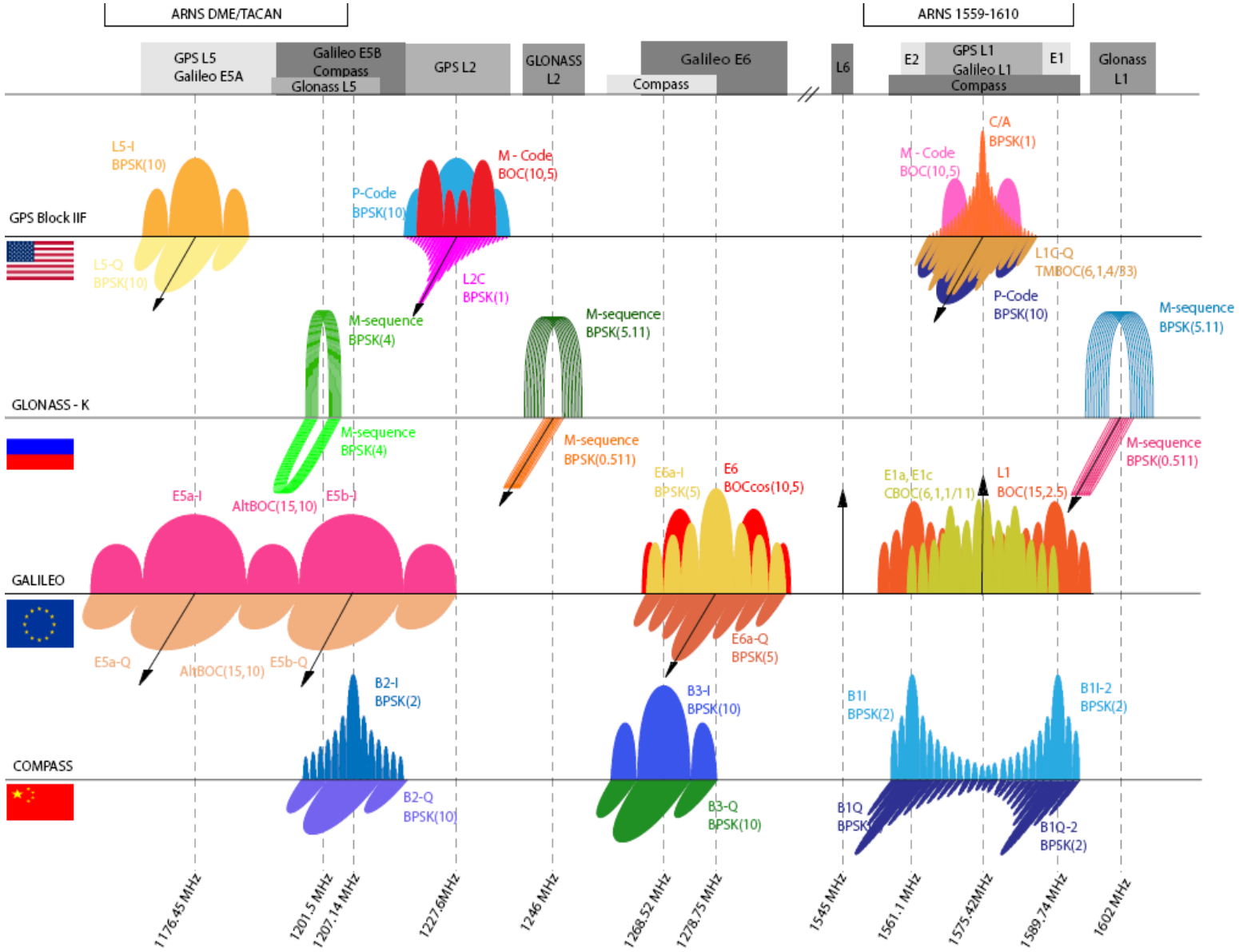
ASEN 5090
Lecture 18
GNSS Spectra



Penny Axelrad

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Where are we?

Fundamentals/Basics

Measurements & Position Solution

Where do measurements come from?

Signals, Receivers Acquisition & Tracking

Applications

Week	Date	Topic	Reading	Assignments - Due 11pm*
1	8/24/26	L1. Introduction & GPS Basics	PNT Ch 1, ME 2.1 & 2.2	
	8/26/26	L2. Navigation Measurements & Methods	ME Ch 1	
	8/28/26			HW1 Assigned
2	8/31/26	L3. GPS Signal Structure	ME Ch 2, PNT Ch 2	
	9/2/26	L4. Coordinate Frames	ME Ch 4.1, 4.A	
	9/4/26			HW1 Due, HW2 Assigned
3	9/7/26	Labor Day (no class)		
	9/9/26	L5. Orbits & Visibility	ME Ch 4.3	
	9/11/26			HW3 Assigned
4	9/14/26	L6. Time Systems, Clocks, & Relativity	ME 4.2, Ashby 2003	
	9/16/26	L7. Measurement Model & IGS Data	ME 5.1, 5.2, https://igs.org/, PNT Ch 27.2	
	9/18/26			HW2 Due, HW4 Assigned
5	9/21/26	L8. Ionospheric Effects & Models	ME Ch 5.3.2, PNT Ch 31	
	9/23/26	L9. Tropospheric Effects and Models	ME Ch 5.3.3	
	9/25/26			HW3 Due
6	9/28/26	L10. Multipath	ME Ch 5.4.2+ PNT Ch 22	
	9/30/26	L11. Error Mitigation & Summary	ME 5.6, 5.8	
	10/2/26			HW4 Due, HW5 Assigned
7	10/5/26	L12. Least Squares Solutions	ME Ch 6.A	
	10/7/26	L13. GPS Point Solutions	ME Ch 6.1	
	10/9/26			
8	10/12/26	L14. Improved GPS Solutions		
	10/14/26	L15. Midterm Review & Homework Discussion		
	10/16/26			HW5 Due, HW6 Assigned
9	10/19/26	L16. Precise Positioning Methods	ME Ch 7	
	10/21/26	Midterm Exam (no lecture)		Midterm Release 10/21 8AM
	10/23/26			Midterm Close 10/25 11PM
10	10/26/26	L17. Introduction to Signals & Spectra	ME Ch 8	
	10/28/26	L18. GNSS Spectrum	ME Ch 9	
	10/30/26			
11	11/2/26	L19. Receiver Overview, Antennas, Link Budget	ME Ch 11, PNT Ch 14	HW6 Due, HW7 Assigned
	11/4/26	L20. GNSS Receiver Architecture	ME Ch 10 (parts)	
	11/6/26			
12	11/9/26	L21. GPS Acquisition	ME Ch 11.2	
	11/11/26	L22. GPS Acquistion (continued)		
	11/13/26			HW7 Due, HW8 Assigned
13	11/16/26	L23. GPS Signal Tracking Basics	ME Ch 12, PNT Ch 15	
	11/18/26	L24. Advanced Signal Tracking (Xu)		
	11/20/26			
14	11/23/26	FALL BREAK		
	11/25/26			
	11/27/26			
15	11/30/26	L25. Space Applications of GNSS	PNT Ch 62-63	
	12/2/26	L26. Aircraft and UAV Applications	PNT Ch 60-61	HW8 Due
	12/4/26	L27. Final Exam Review (Monday Class Schedule)		
16	12/7/26	Final Exam available starting Friday, 12/4		Final Release 12/4 6PM
	12/9/26	Final Exam due Tues, 12/8 (24hr time limit)		Final Close 12/8 11PM
	12/11/26			

Outline

- Review Fourier series for periodic functions
- Fourier transforms for pulse & frequency shift
- Spectrum of a PRN code & GPS legacy signals
- Modernized signals – BOC

Periodic Functions

- A periodic function of period T_p = 1/fp

$$f\left(t+nT_p\right)=f\left(t\right)$$

- Can be expressed by a Fourier Trigonometric Series as:

$$f\left(t\right)=a_0+\sum_{n=1}^{\infty}\left[a_n\cos n\left(2\pi f_p t\right)+b_n\sin n\left(2\pi f_p t\right)\right]$$

Synthesis equation – shows how f(t) can be reconstructed by adding up an infinite number of sines & cosines

Fourier series for a signal that is periodic in P

$$f(t) = a_0 + \sum_{n=1}^{\infty}\left[a_n \cos n\left(2\pi f_p t\right) + b_n \sin n\left(2\pi f_p t\right)\right]$$

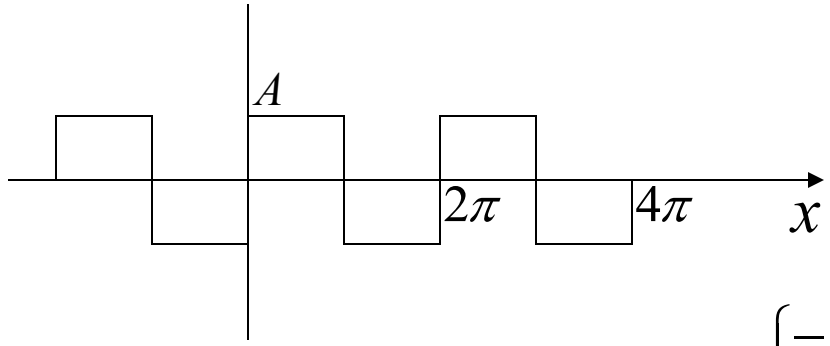
analysis equations – show how to find the amount of each sine and cosine wave that is needed.

$$a_0 = \frac{1}{P}\int_{-P/2}^{P/2} f(t)\,dt$$

$$a_n = \frac{1}{P/2}\int_{-P/2}^{P/2} f(t)\cos n2\pi f_p t\;dt$$

$$b_n = \frac{1}{P/2}\int_{-P/2}^{P/2} f(t)\sin n2\pi f_p t\;dt$$

Square Wave



$$f(x) = \begin{cases} -A & \text{if } -\pi < x < 0 \\ A & \text{if } 0 < x < \pi \end{cases}$$

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{0} -A\,dx + \frac{1}{2\pi}\int_{0}^{\pi} A\,dx = 0$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{0} -A\cos nx\;dx + \frac{1}{\pi}\int_{0}^{\pi} A\cos nx\;dx = 0 \text{ for all } n$$

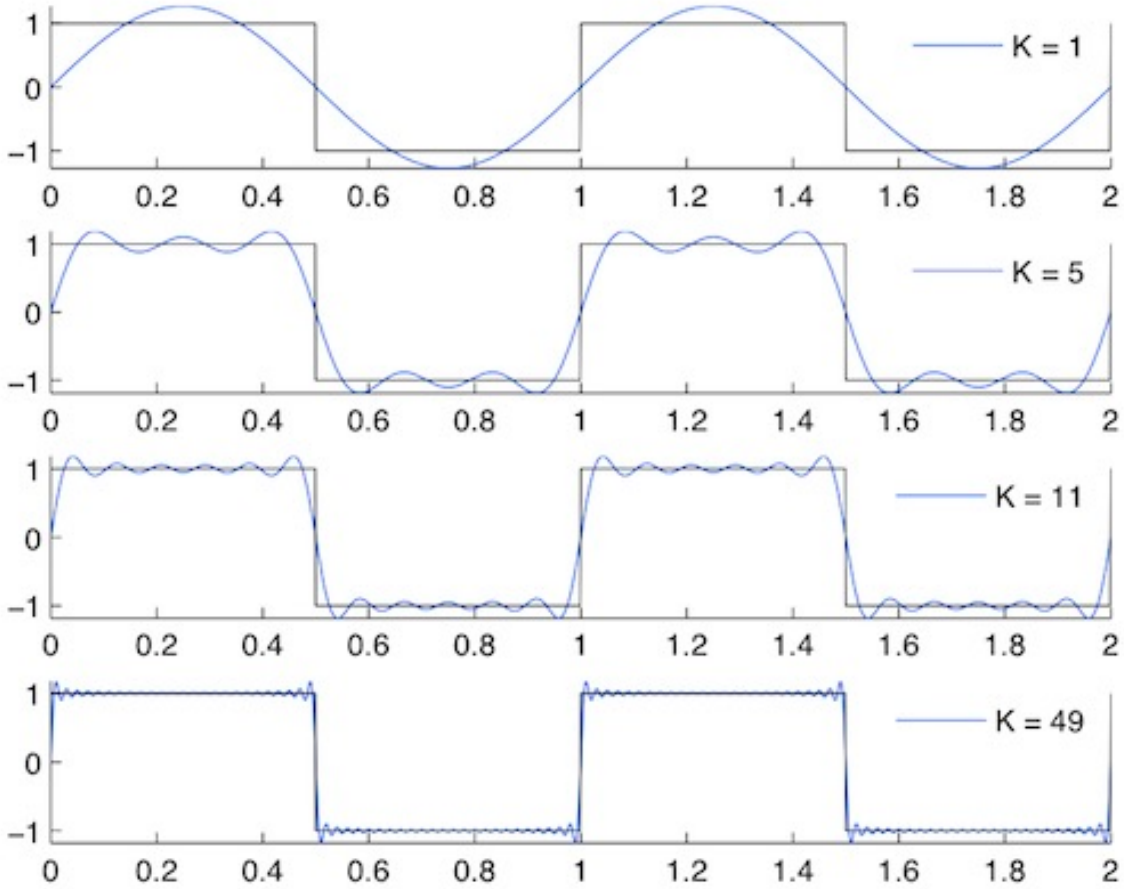
$$b_n = \frac{1}{\pi}\int_{-\pi}^{0} -A\sin nx\;dx + \frac{1}{\pi}\int_{0}^{\pi} A\sin nx\;dx = \frac{2A}{n\pi}\left(1-\cos n\pi\right)$$

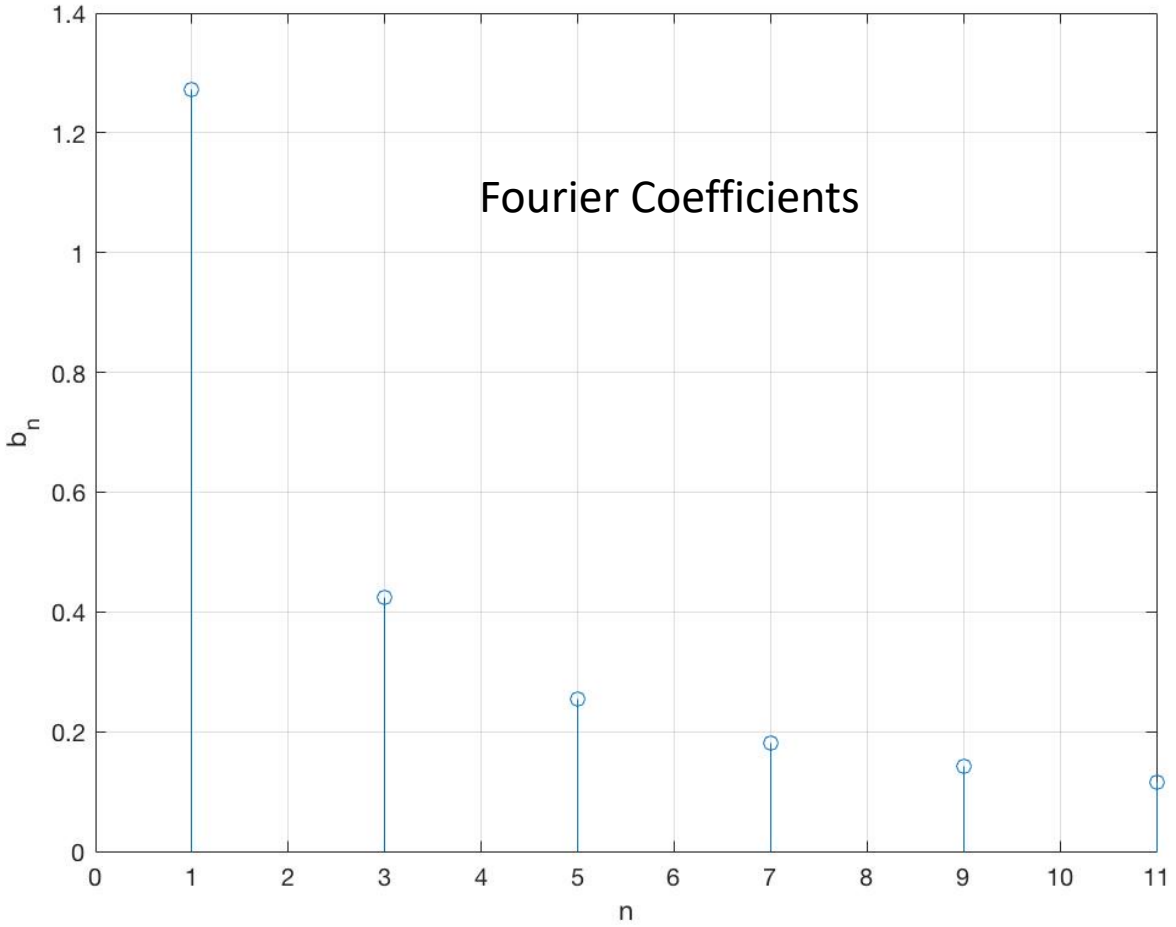
$$b_1 = \frac{4A}{\pi}\;,\; b_2 = 0\;,\; b_3 = \frac{4A}{3\pi}\;,\; b_4 = 0\;,etc.$$

$$\text{so}\quad f(x) = \frac{4A}{\pi}\left[\sin x + \tfrac{1}{3}\sin 3x + \tfrac{1}{5}\sin 5x + \ldots\right]$$

Reconstructed function based on the Fourier series

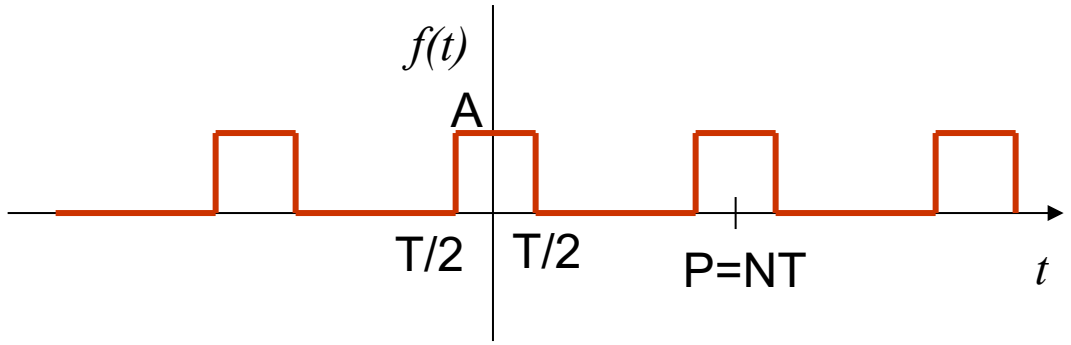
Fourier Series Approximations





Periodic rectangular pulse

$$f(t) = \begin{cases} 0 & \text{if } |t| > \frac{T}{2} \\ A & \text{if } |t| < \frac{T}{2} \end{cases}$$



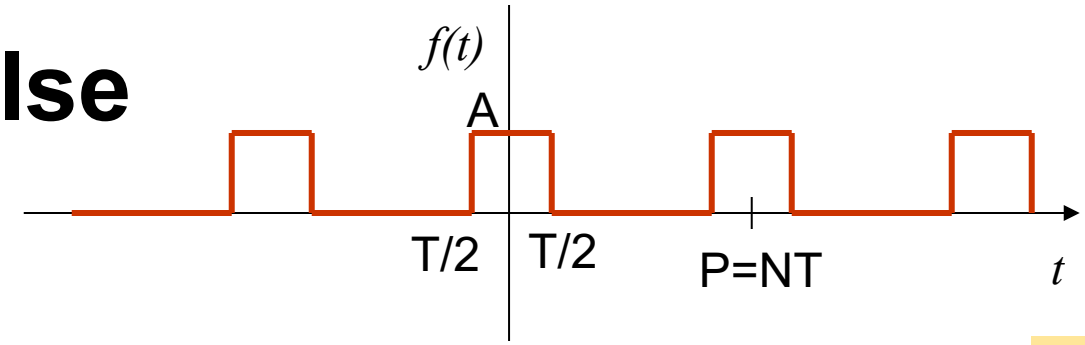
The textbook refers to this a square wave in Fig 8.12.

$$P = NT = \frac{1}{f_P}$$

How can we express as a Fourier trig series?

$$f(t) = a_0 + \sum_{n=1}^{\infty}\left[a_n \cos n\left(2\pi f_p t\right) + b_n \sin n\left(2\pi f_p t\right)\right]$$

Periodic rectangular pulse



What is T?
What is P?

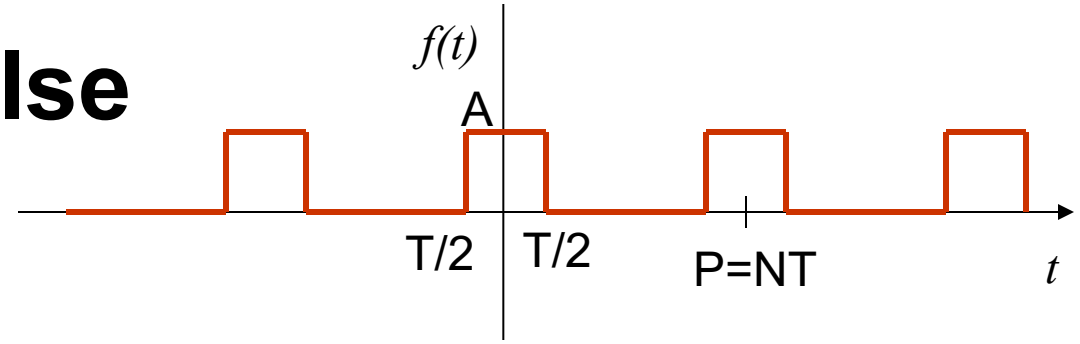
$$f(t) = \begin{cases} 0 & \text{if } |t| > \frac{T}{2} \\ A & \text{if } |t| < \frac{T}{2} \end{cases}$$

$$P = NT = \frac{1}{f_P}$$

$$a_0 = \frac{1}{P}\int_{-P/2}^{P/2} f(t)\,dt = \frac{1}{P}\int_{-T/2}^{T/2} A\,dt = \frac{AT}{P} = \frac{A}{N}$$

"DC term"

Periodic rectangular pulse



$$f(t) = \begin{cases} 0 & \text{if } |t| > \frac{T}{2} \\ A & \text{if } |t| < \frac{T}{2} \end{cases} \qquad P = NT = \frac{1}{f_P}$$

$$a_n = \frac{1}{P/2}\int_{-P/2}^{P/2} f(t)\cos n2\pi f_p t \; dt = \frac{1}{P/2}\int_{-T/2}^{T/2} A\cos n2\pi f_p t \; dt = \frac{A\sin n2\pi f_p t}{n2\pi f_p P/2}\Bigg|_{T/2}^{T/2}$$

How does this compare to square wave?

$$a_n = \frac{2A\sin n\pi f_p T}{n\pi} = \frac{2A}{N}\frac{\sin n\pi/N}{n\pi/N} = \frac{2A}{N}\operatorname{sinc}\frac{n\pi}{N}$$

$$a_1 = \frac{2A}{N}\frac{\sin \pi/N}{\pi/N}, a_2 = \frac{2A}{N}\frac{\sin 2\pi/N}{2\pi/N}, \ldots$$

Remember what these coefficients mean:

$$f(t) = a_0 + \sum_{n=1}^{\infty}\left[a_n \cos n\left(2\pi f_p t\right) + b_n \sin n\left(2\pi f_p t\right)\right]$$

Periodic rectangular pulse (continued)

For what values of "n" is a_n=0? (what multiples of the repeat frequency?)

$$a_n = \frac{2A}{N}\frac{\sin n\pi / N}{n\pi / N} = 0 \text{ for } \quad \frac{n}{N} = \text{is a non-zero integer}$$

$$n = N, 2N, 3N, etc$$

$$Nf_P = \frac{1}{T} = f_C$$

So, for $Nf_P t,\ 2Nf_P t,\ 3Nf_P t,\ 4Nf_P t$

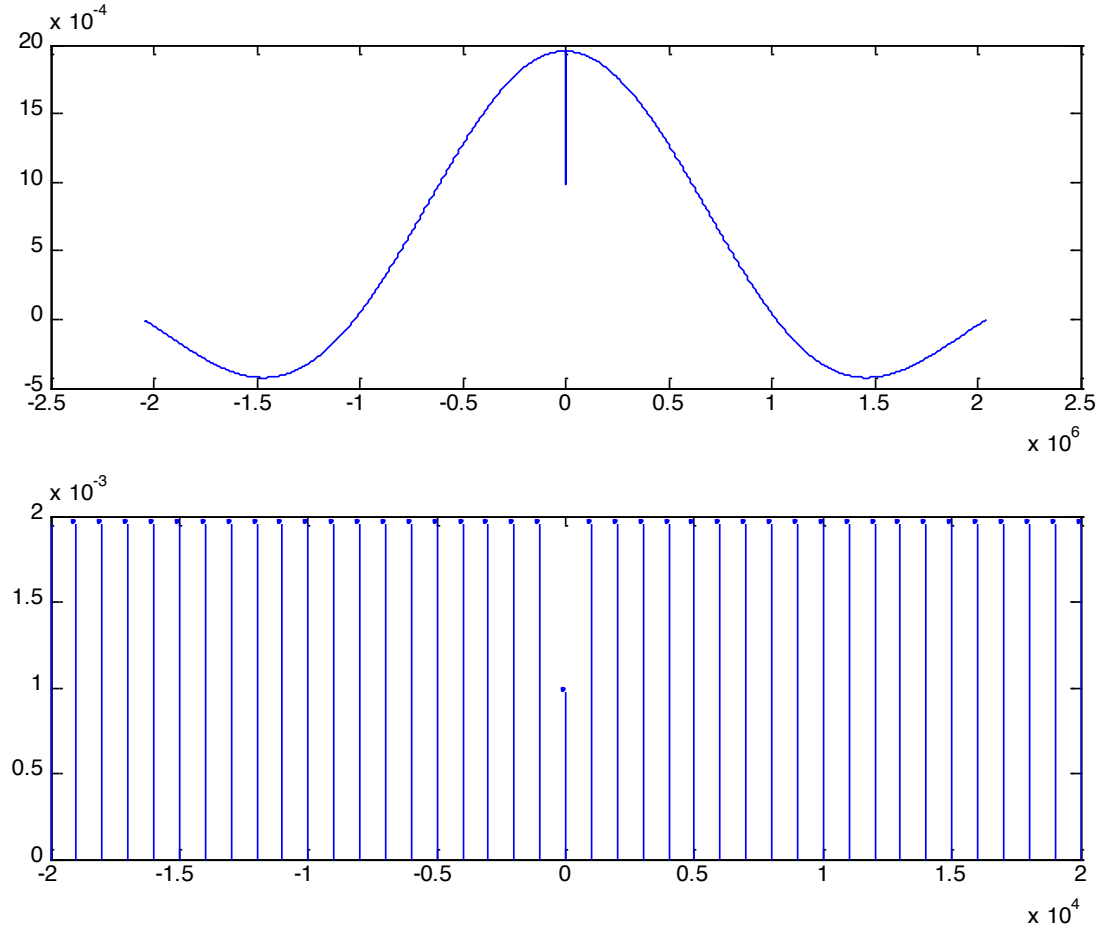
and $\quad -Nf_P t,\ -2Nf_P t,\ -3Nf_P t,\ -4Nf_P t$

$$f(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\left(2\pi f_p t\right)$$

Fourier representation of a periodic rectangular pulse

$$f(t) = \frac{A}{N} + \frac{2A}{N}\sum_{n=1}^{\infty}\left[\operatorname{sinc}\frac{n\pi}{N}\cos n\left(2\pi f_p t\right)\right]$$

a_n



Series for 1/T=1.023e6 and N = 1023

Repeat period of signal = 0.001 sec

f_P = 1000 Hz

Zoomed in to show individual values

FOURIER TRANSFORM

- For any $x(t)$ you can take a Fourier Transform, to produce a continuous function of frequency:

$$X(f) = \mathcal{F}\{x(t)\} = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}dt$$

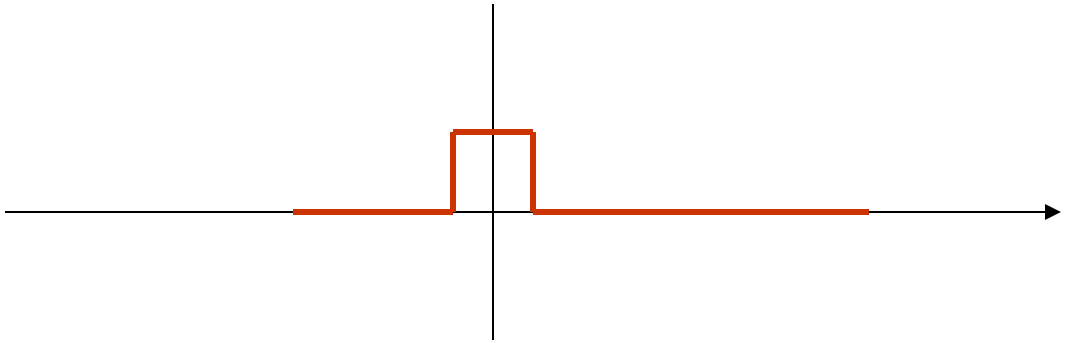
- To go from frequency domain to time domain use the inverse transform:

$$x(t) = \mathcal{F}^{-1}\{X(f)\} = \int_{-\infty}^{\infty} X(f)e^{j2\pi ft}df$$

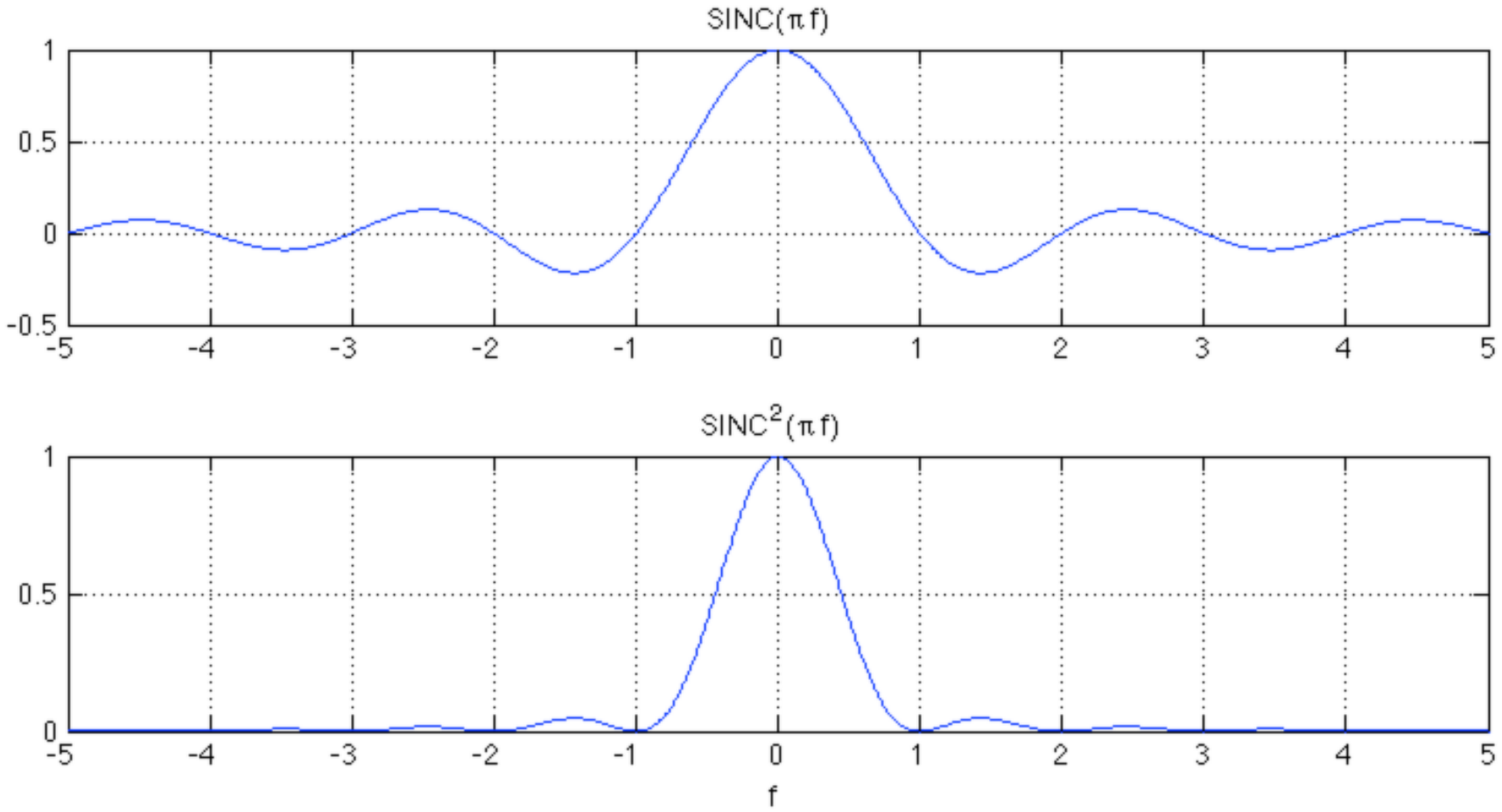
Table 8.2 Fourier transform pairs

Time function	*Fourier transform*
Unit pulse $p(t)$	$\text{sinc}(\pi f)$
Shifted and scaled pulse $Ap\left(\frac{t-\tau}{T_C}\right)$	$AT\exp(-j2\pi f\tau)\text{sinc}(\pi fT)$
Unit impulse function $\delta(t)$	1
$\delta(t-t_0)$	$\exp(-j2\pi ft_0)$
1	$\delta(f)$
$\exp(j2\pi f_0 t)$	$\delta(f-f_0)$
$\cos(2\pi f_0 t)$	$\frac{1}{2}\delta(f-f_0)+\frac{1}{2}\delta(f+f_0)$
$\sin(2\pi f_0 t)$	$\frac{1}{2j}\delta(f-f_0)-\frac{1}{2j}\delta(f+f_0)$
$\text{sgn}\,t=\begin{cases}1 & t\geq 0\\ -1 & t<0\end{cases}$	$\frac{1}{j\pi f}$
Step function $u(t)$	$\frac{1}{2}\delta(f)+\frac{1}{j2\pi f}$
Ramp function $tu(t)$	$j\pi\dot{\delta}(f)-\frac{1}{(2\pi f)^2}$
Comb function $\sum_{i=-\infty}^{\infty}\delta(t-iT_S)$	$\frac{1}{T_S}\sum_{i=-\infty}^{\infty}\delta\left(f-\frac{i}{T_S}\right)$

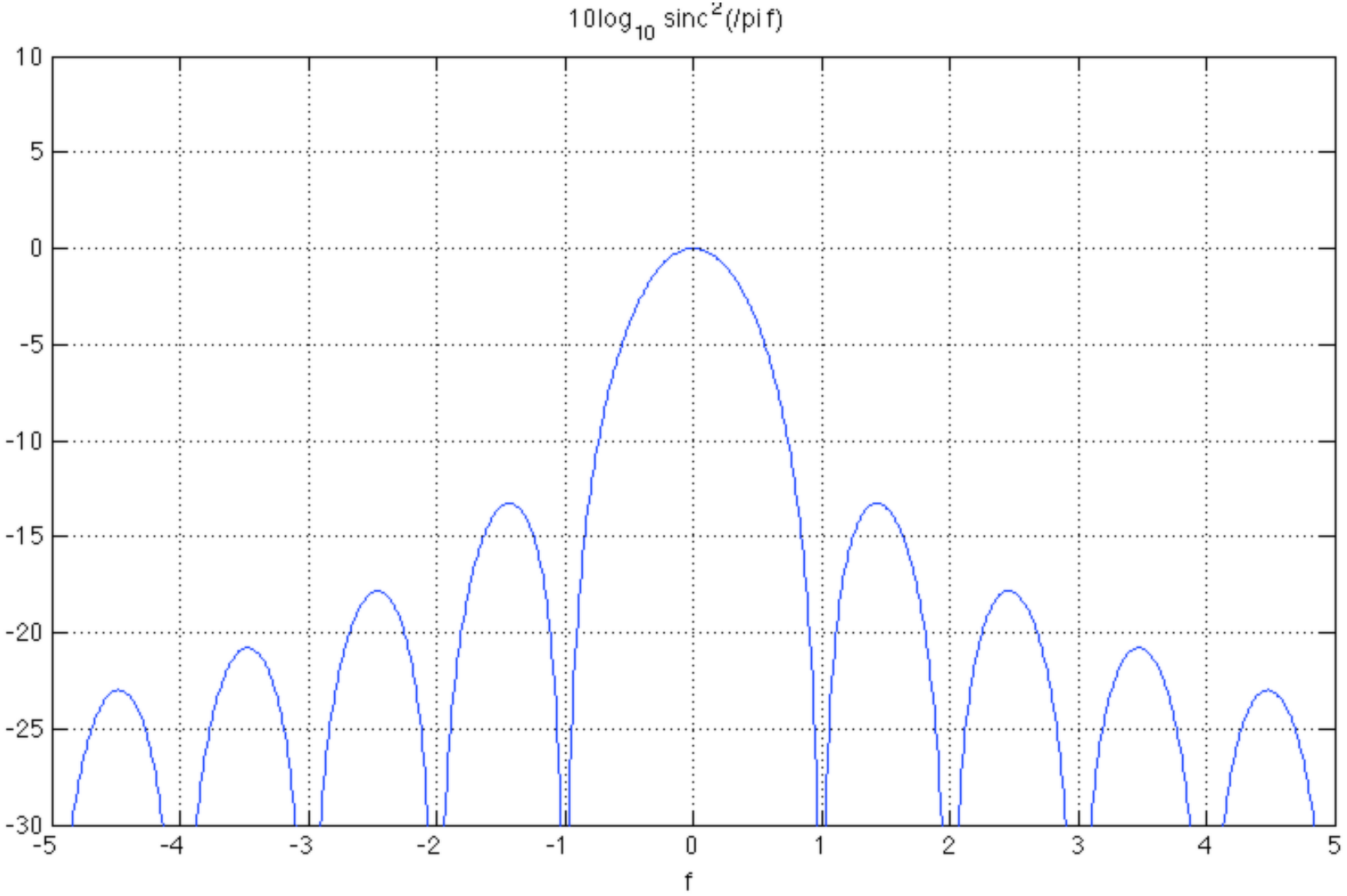
Fourier Transform of a Unit Pulse



Sinc and Sinc^2



sinc^2 in dB



Fourier Transform Properties

Table 8.1 Properties of Fourier transforms

Property/operation	*Time function*	*Fourier transform*
Linearity	$k_1 x(t) + k_2 y(t)$	$k_1 X(f) + k_2 Y(f)$
Time delay	$x(t - t_0)$	$X(f)\exp(-j2\pi f t_0)$
Time scaling	$x(kt)$	$\frac{1}{\lvert k\rvert} X\left(\frac{f}{k}\right)$
Differentiation	$\frac{dx}{dt}$	$j2\pi f X(f)$
n differentiations	$\frac{d^n x}{dt^n}$	$(j2\pi f)^n X(f)$
Symmetry	$\mathcal{F}\{x(t)\} = X(f)$	$\mathcal{F}\{X(t)\} = x(-f)$
Frequency translation	$x(t)\exp(j2\pi f_0 t)$	$X(f - f_0)$
Modulation	$x(t)\cos(j2\pi f_0 t)$	$\frac{1}{2} X(f - f_0) + \frac{1}{2} X(f + f_0)$
Modulation	$x(t)\sin(j2\pi f_0 t)$	$\frac{1}{2j} X(f - f_0) - \frac{1}{2j} X(f + f_0)$
Convolution	$x(t) * h(t)$	$X(f)H(f)$
Multiplication	$x(t) \times h(t)$	$X(f) * H(f)$
Integration	$\int_{-\infty}^{t} x(\varphi)\, d\varphi$	$\frac{X(f)}{j2\pi f} + \frac{1}{2} X(0)\delta(f)$

Fourier Transform Frequency Shift

$$\Im\{a(t)\cos 2\pi f_0 t\} = \frac{1}{2}A(f - f_0) + \frac{1}{2}A(f + f_0) \qquad (9.3)$$

Start with a signal centered on f_{L1}

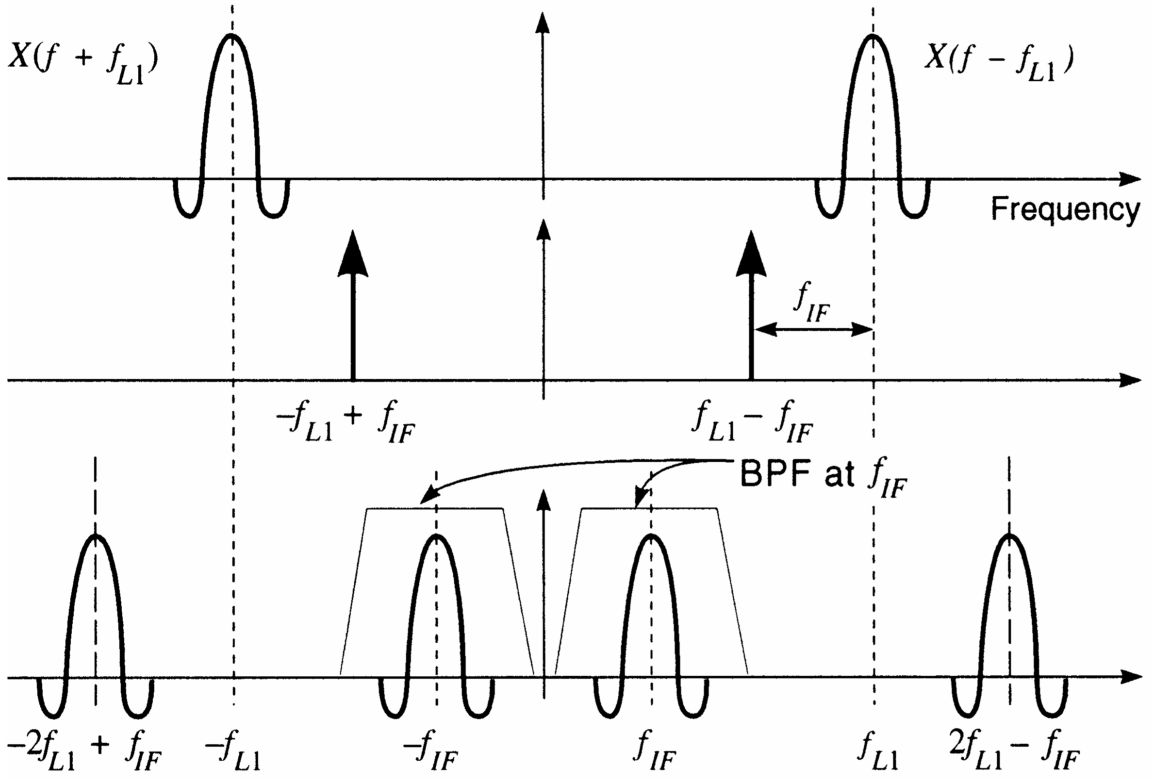
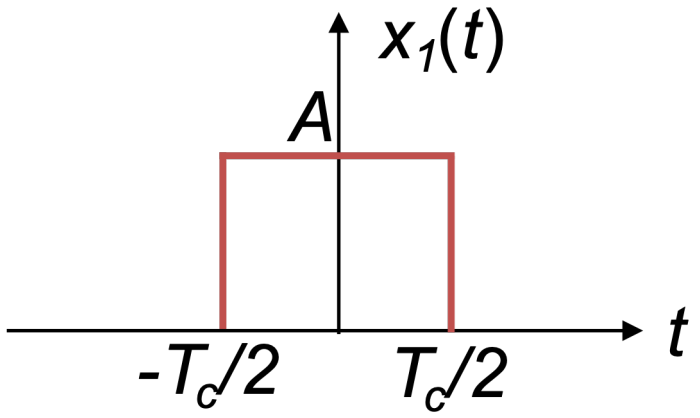


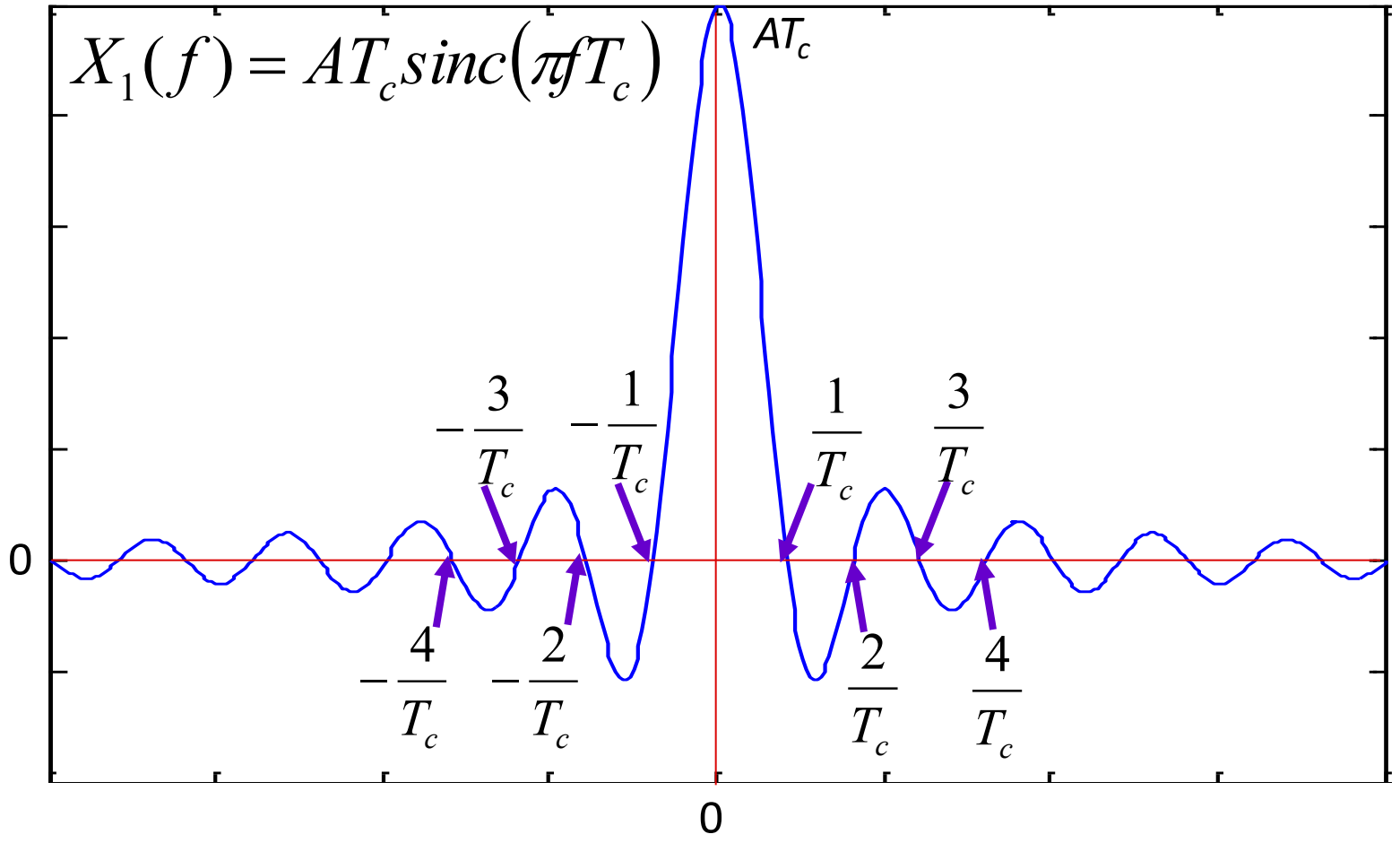
Figure 9.6 Fourier transforms before and after down conversion.

Example 1: Rectangle Pulse Signal Spectrum

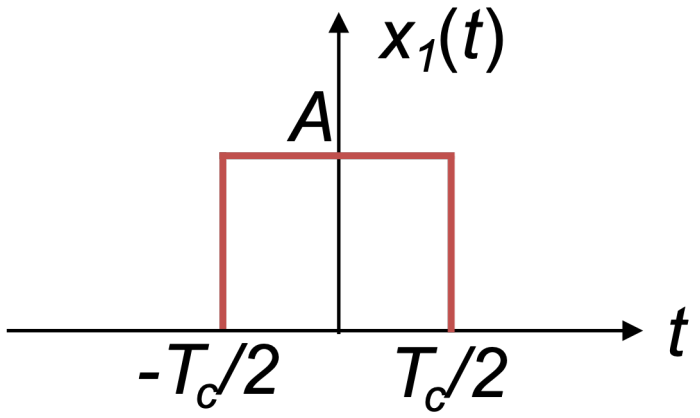


$$X_1(f) = AT_c \frac{\sin(\pi f T_c)}{\pi f T_c}$$

$$sinc(\pi f T_c)$$

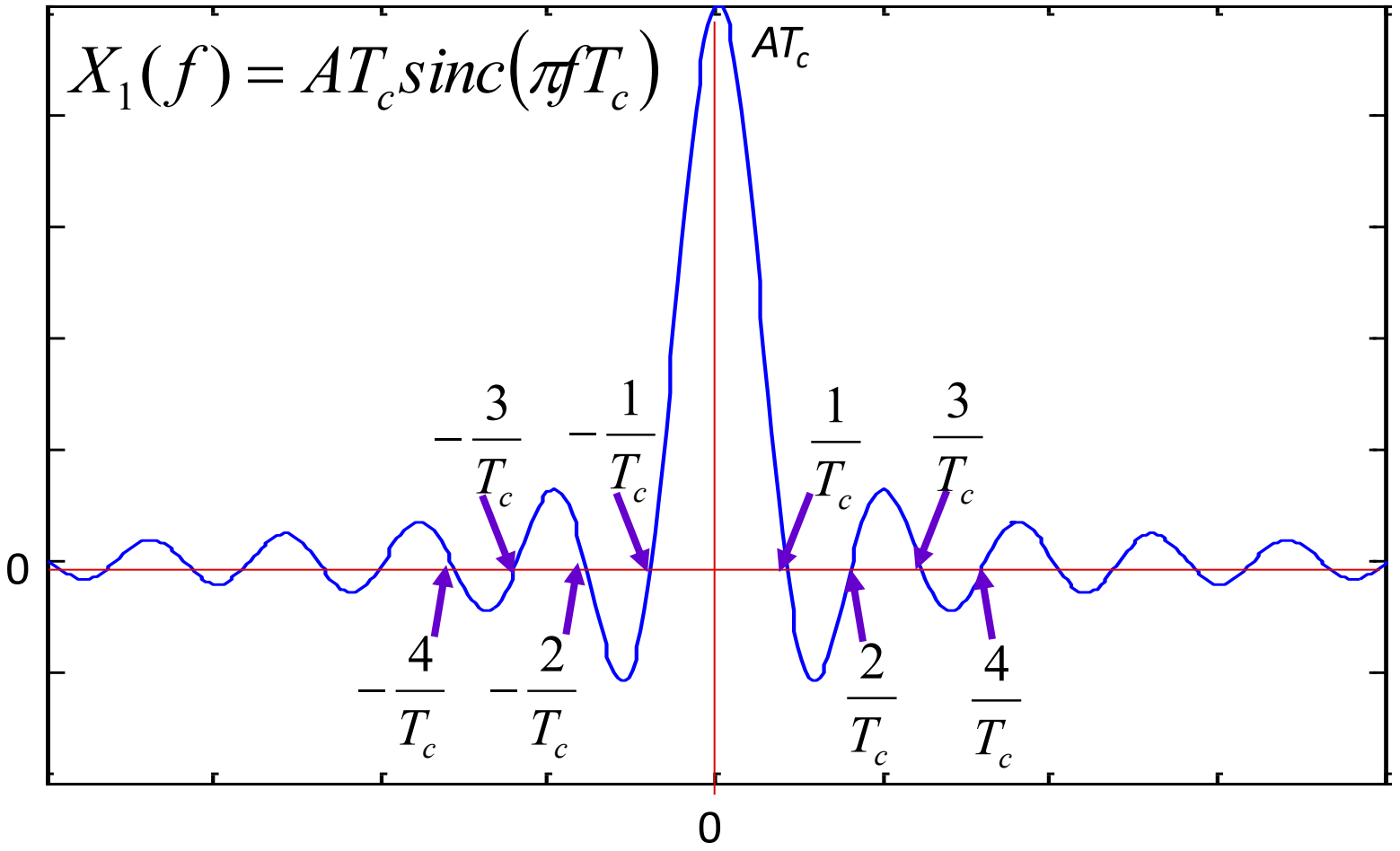


Example 1: Rectangle Pulse Signal Spectrum



$$X_1(f) = AT_c \frac{\sin(\pi f T_c)}{\pi f T_c}$$

$$\mathrm{sinc}(\pi f T_c)$$



PRN CODE

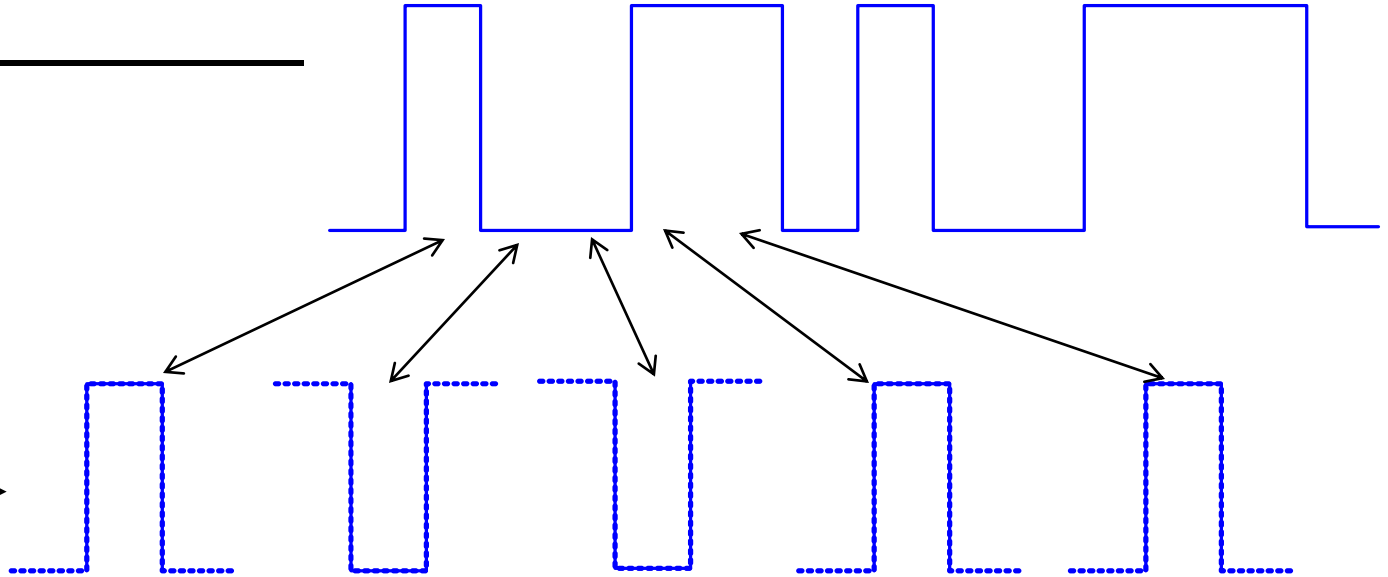
- PRN Code is a sequence of pulses

$$x(t) = \sum_{i=0}^{N-1} x_i p\left(\frac{t - iT_c}{T_c}\right) \qquad x(t) = p\left(\frac{t_c}{T_c}\right) * \sum_{i=0}^{N-1} x_i \delta(t - iT)$$

- If X(f) is Fourier Transform of the code sequence

Example 2: Random Binary Code Sequence

Can be decomposed into
a sequence of rectangle pulses



A rectangle pulse auto-correlation:

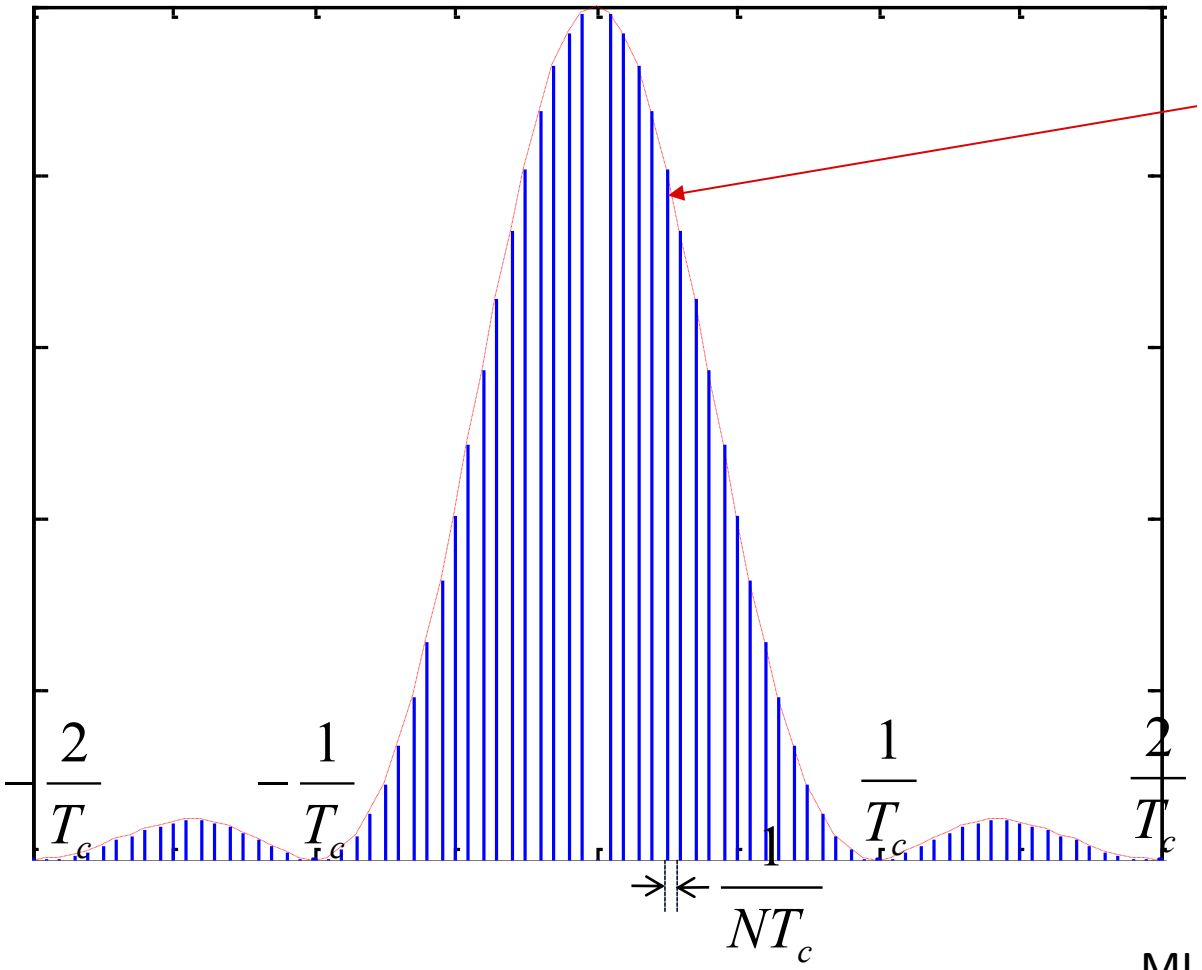
$$R_1(\tau) = \begin{cases} A^2 T_c \left(1 - \dfrac{|\tau|}{T_c}\right) & |\tau| \le T_c \\ 0 & |\tau| \ge T_c \end{cases}$$

A sequence of N pulses correlation:

$R_2(\tau) = R_1(\tau)$N

$$R_2(\tau) = \begin{cases} A^2 N T_c \left(1 - \dfrac{\tau}{T_c}\right) & |\tau| \le T_c \\ 0 & |\tau| \ge T_c \end{cases}$$

Example 3: Maximum Length Sequence (MLS) Power Spectrum

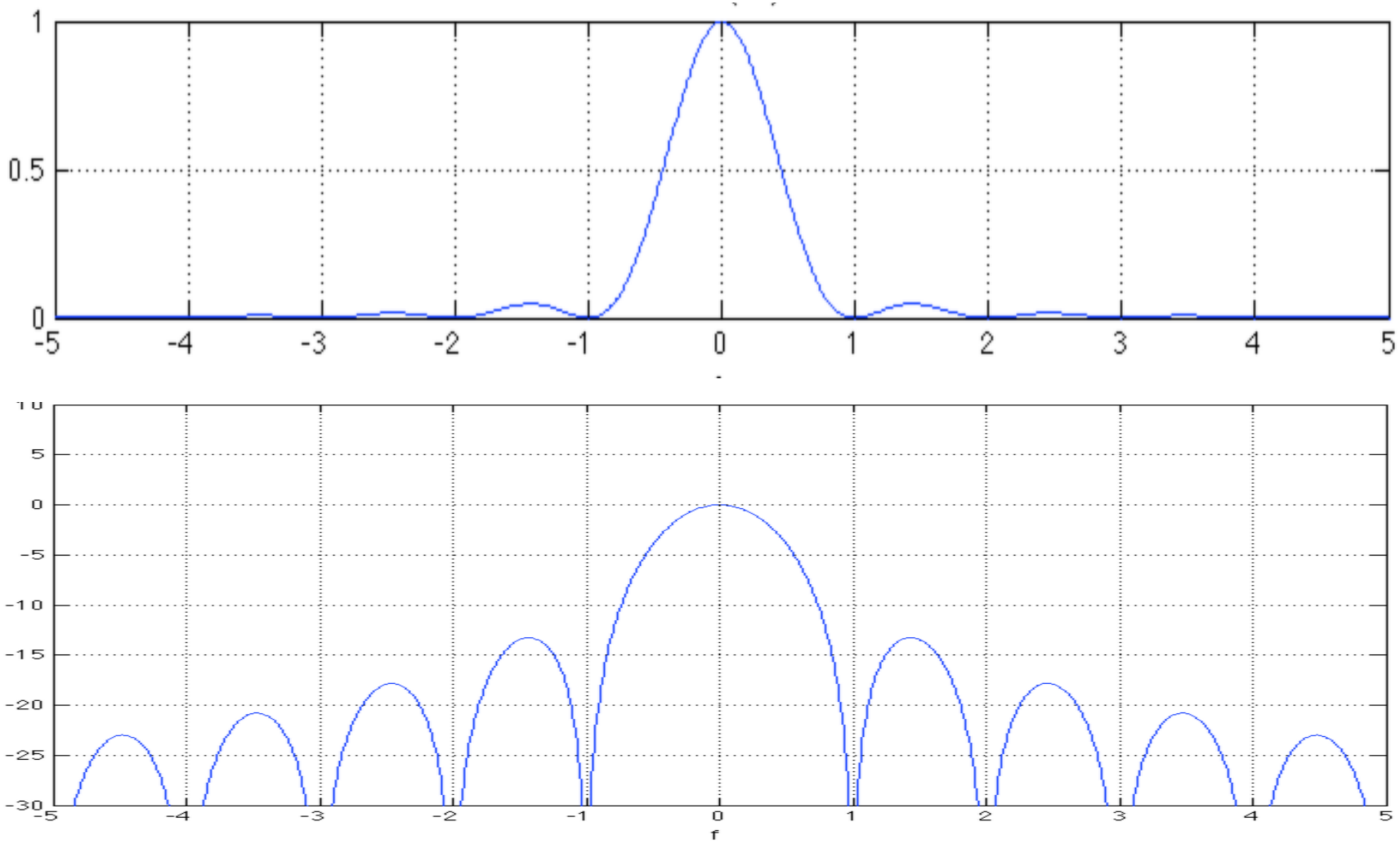


Envelope

$$S_2(\omega) = A^2 N T_c^2 sinc^2(\pi f T_c)$$

MLS is NOT a completely ”random” sequence: there is an extra “1”

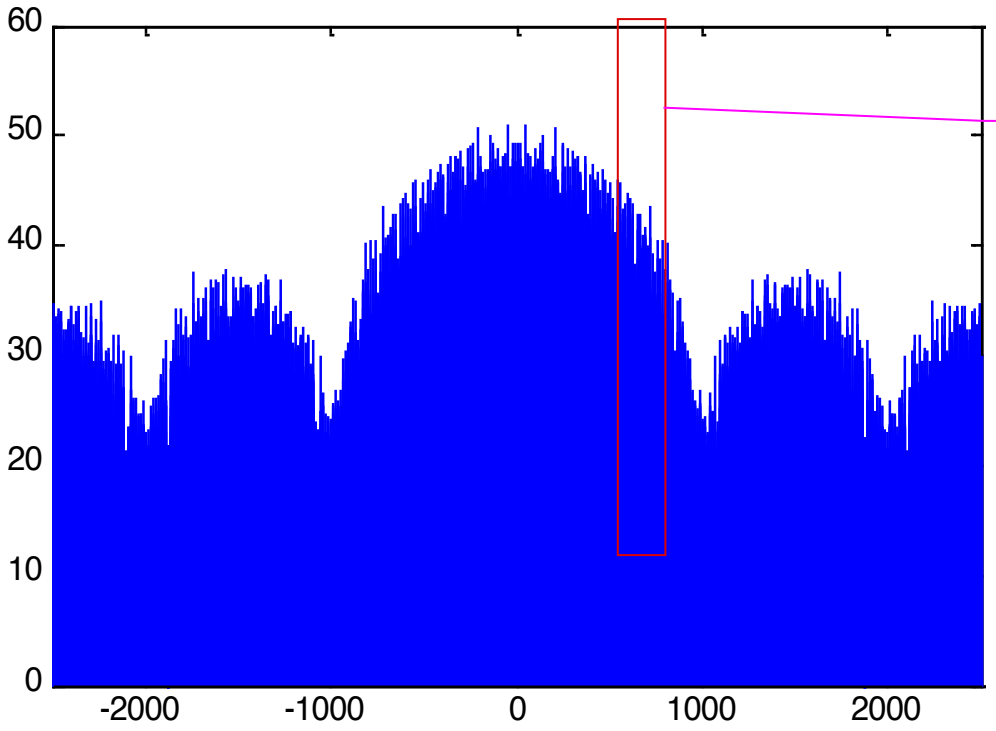
sinc^2 and sinc^2 (dB)

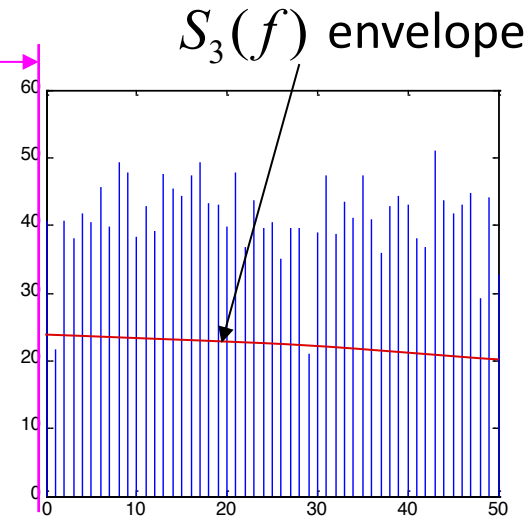


What does this look like for C/A or P(Y) code?

$$F(f) = \frac{1}{f_C}\,\mathrm{sinc}\left(\frac{\pi f}{f_C}\right)X(f) = \frac{1}{f_C}\,\frac{\sin\left(\pi f / f_C\right)}{\pi f / f_C}X(f)$$

Example 4: GPS CA Code Power Spectrum (in dB)





Signal parameters:

fs=5 MHz;	*%Sample rate*
A=1;	*%Amplitude*
T=1 ms;	*%Period*
L=1 ms;	*%Data length*

Now include a carrier signal – what does the spectrum look like?

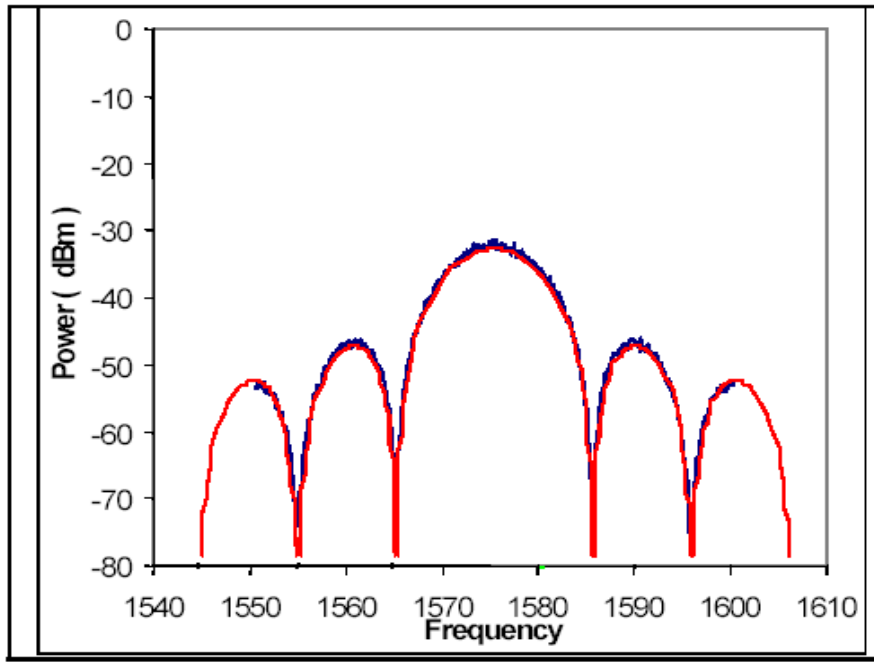
$$s(t) = x(t)\cos(2\pi f_{L1} t)$$

$$S(f) = F\{s(t)\}$$

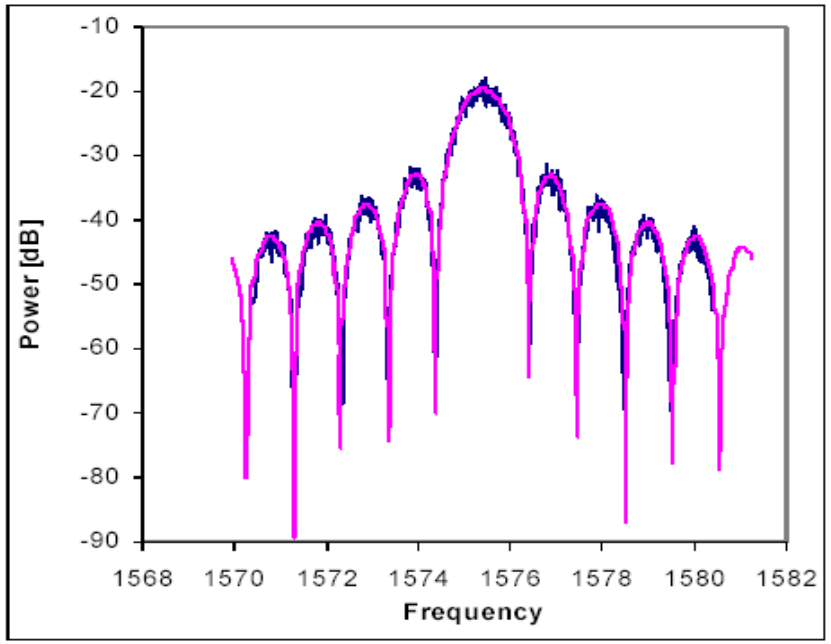
PRN CODE MODULATED SIGNAL

- Dominant feature is the sinc function, with the center shifted up to the carrier frequency

P(Y) Code on L1



C/A Code on L1



Transmitted Signals

$$s_{L1}(t) = \sqrt{2P_{C1}}D(t)x(t)\cos(2\pi f_{L1}t + \theta_{L1}) + \sqrt{2P_{Y1}}D(t)y(t)\sin(2\pi f_{L1}t + \theta_{L1})$$

$$s_{L2}(t) = \sqrt{2P_{Y2}}D(t)y(t)\sin(2\pi f_{L2}t + \theta_{L2})$$

$D(t)$ is navigation data

$x(t)$ is C/A code, $y(t)$ is P(Y) code

P is signal power

$f_{L1} = 1575.42MHz,\ \lambda_{L1} \approx 19cm,\ f_{L2} = 1127.6MHz,\ \lambda_{L2} \approx 24cm$

$f_{C/A} = 1.023MHz,\ T_{C/A} \approx 0.996\mu s,\ \lambda_{C/A} = 293m$

$f_{data} = 50Hz,\ T_{data} = 20ms$

What are the characteristics and effects of the specific code used?

MAXIMAL LENGTH CODES

- Maximal length codes are the longest codes (i.e. sequence of 1’s and 0’s) that can be generated by a given number of shift register stages. For an *n*-stage shift register, the longest code is 2^n - 1 chips (i.e. 1's or 0’s). Maximal length codes have many interesting properties, especially their autocorrelation function.
- For a maximal length code, the autocorrelation function for a phase shift other than 0 or (2^n - 1) is -1.
- For any code, the autocorrelation for a 0 shift is equal to the number of chips in the code.

Gold Codes

- Pair of maximal length sequences ($N = 2^n-1$) can be used to generate a family of N+2 Gold sequences for n not divisible by 4:
- n=3:N=7, n=5:N=31, n=6:N=63
- Autocorrelation has only 4 possible values:

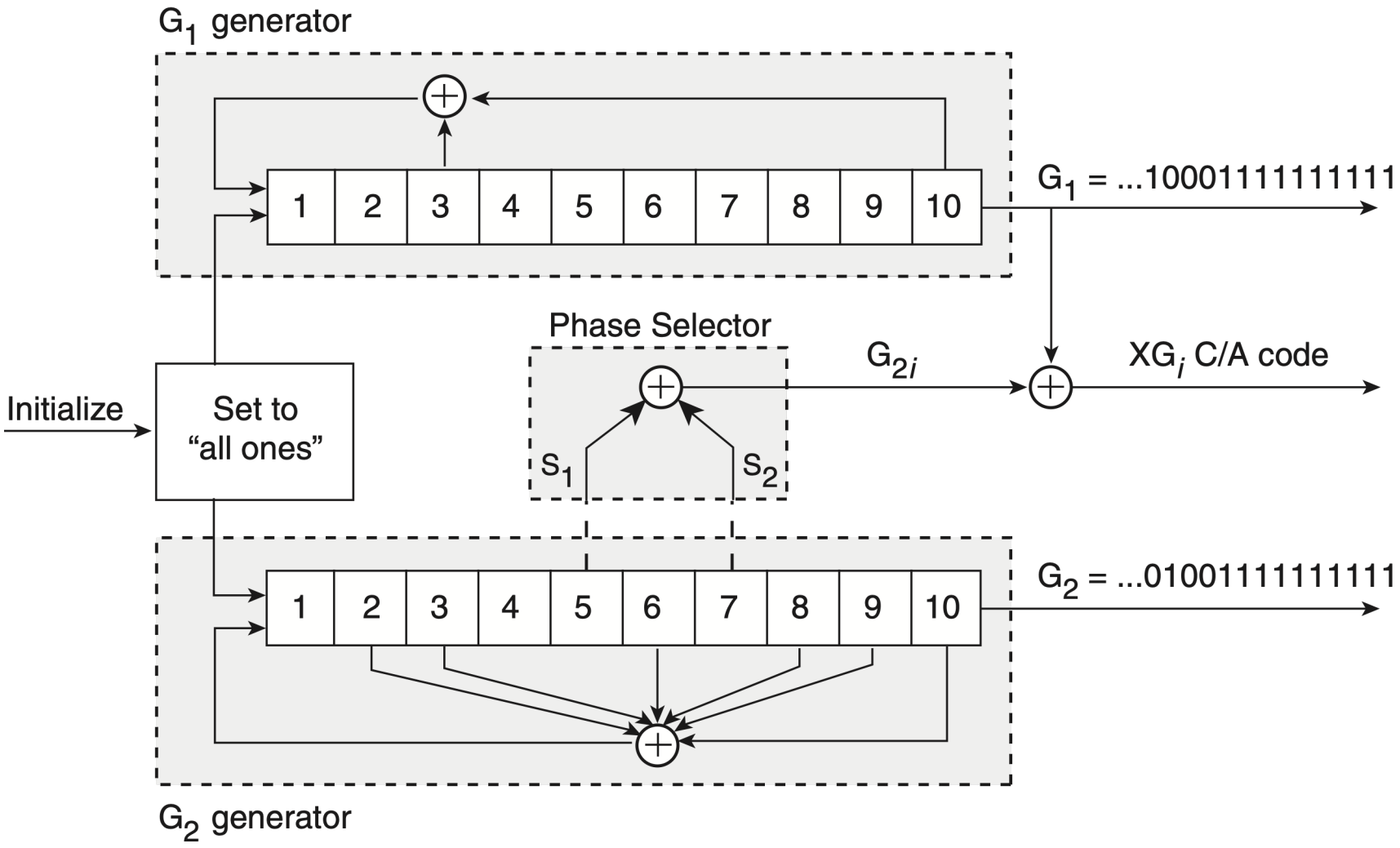
$$R(\tau) \in \left\{ \quad 1, \qquad -\frac{1}{N}, \qquad -\frac{\beta}{N}, \qquad \frac{\beta-2}{N} \quad \right\}$$

$$\text{where } \beta = 1+2^{\lfloor (n+2)/2 \rfloor}$$

- For C/A code n=10, N=1023, β=65

$$R(\tau) \in \left\{ 1, \qquad -\frac{1}{1023}, \qquad -\frac{65}{1023}, \qquad \frac{63}{1023} \right\}$$

C/A Code Generator



AUTOCORRELATION

$$R(\tau) = \frac{1}{T_{code}} \int_0^{T_{code}} x(t)x(t-\tau)\,dt$$

for

$$\tau = nT_c \quad R(nT_c) = \frac{1}{N}(agree - disagree)$$

within a chip

$R(\tau)$ changes linearly from $R(iT_c)$ to $R((i+1)T_c)$

$$R(\tau) = \begin{cases} \frac{\tau}{T_c} + 1 & \text{for } -T_c < \tau < 0 \\ -\frac{\tau}{T_c} + 1 & \text{for } 0 < \tau < T_c \\ 0 & otherwise \end{cases}$$

Power Spectrum and Auto-correlation

Power spectrum $$S(f) = |X(f)|^2$$

Auto-correlation $$R(\tau) = \int_{-\infty}^{\infty} x(t)x(t+\tau)dt$$

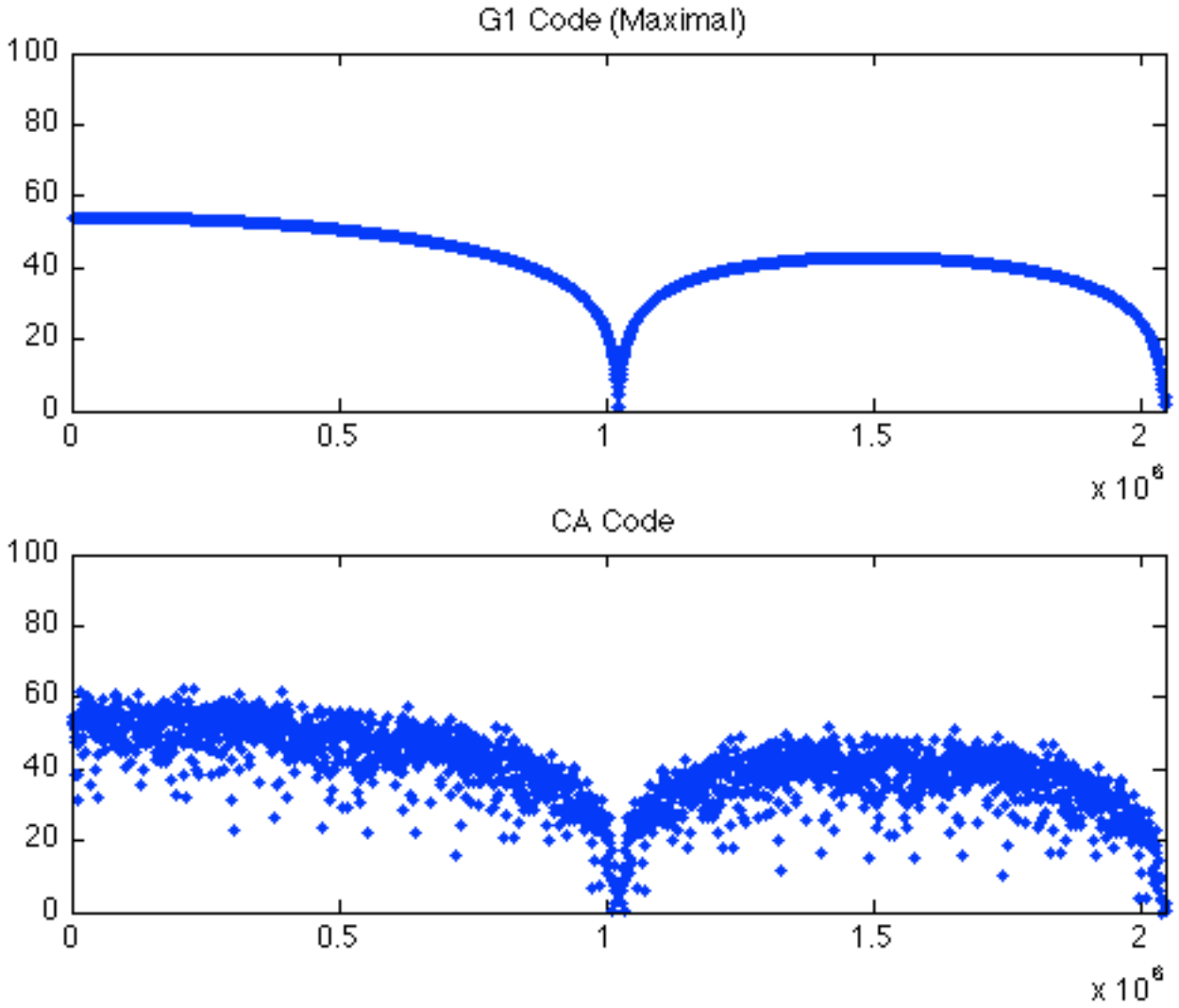
If $x(t)$ is real, $$S(f) = \int_{-\infty}^{\infty} R(\tau)e^{-j2\pi f\tau}d\tau$$

$$R(\tau) \overset{FT}{\Leftrightarrow} S(f)$$

A signal's auto-correlation and power spectrum are a FT pair

For PROBLEM 2-2

- Check autocorrelation and cross correlation values for codes you generated.
- Compute FFT



```
% Example for lecture
fs = 50e6; Ts = 1/fs; % Sample frequency and time interval between samples
fc = 1.023e6; Tc = 1/fc; % C/A code chipping rate in Hz

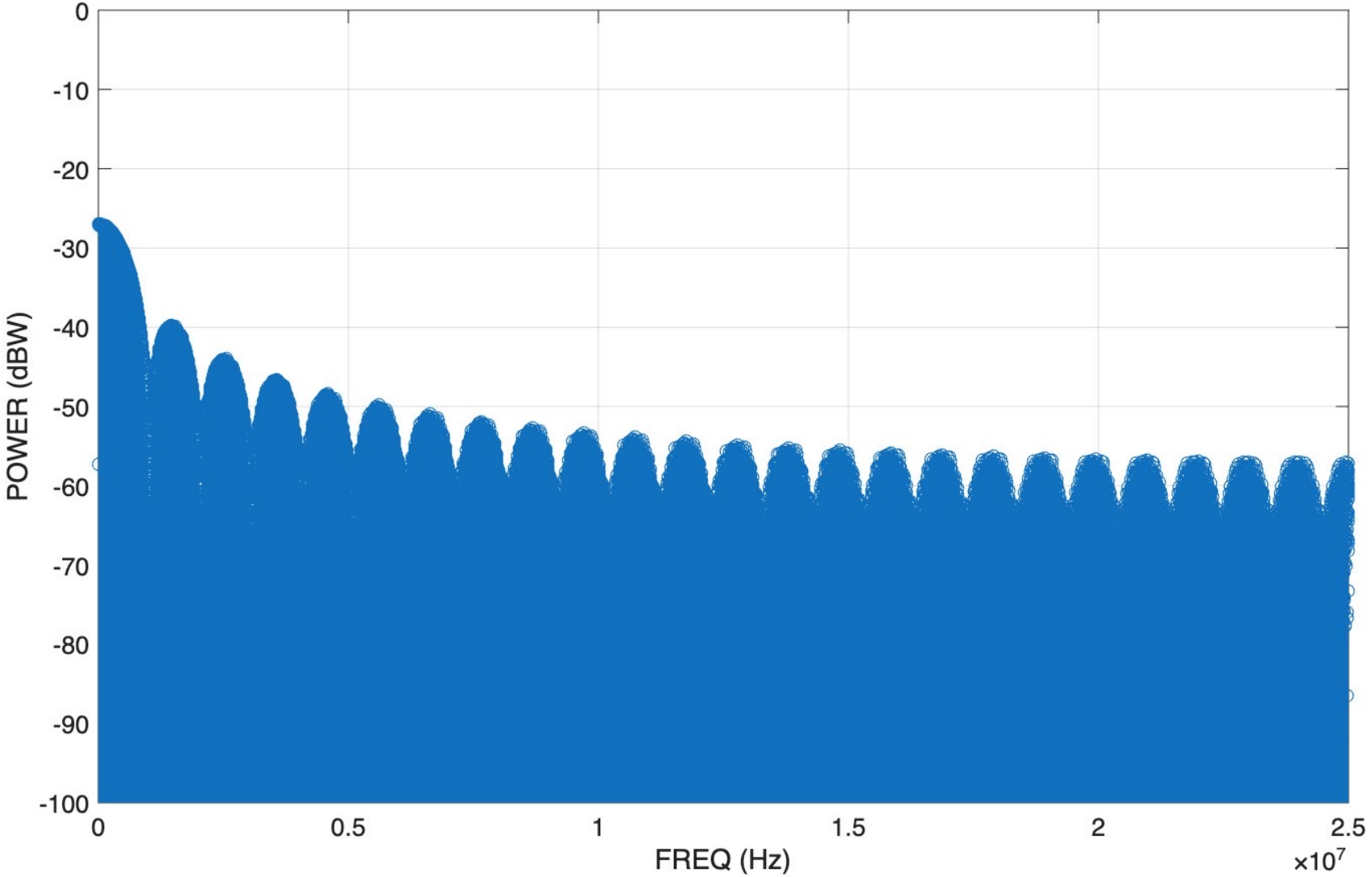
tf = 5e-3; % 5 ms
npts = tf/Ts;
t = Ts*[0:npts-1]';

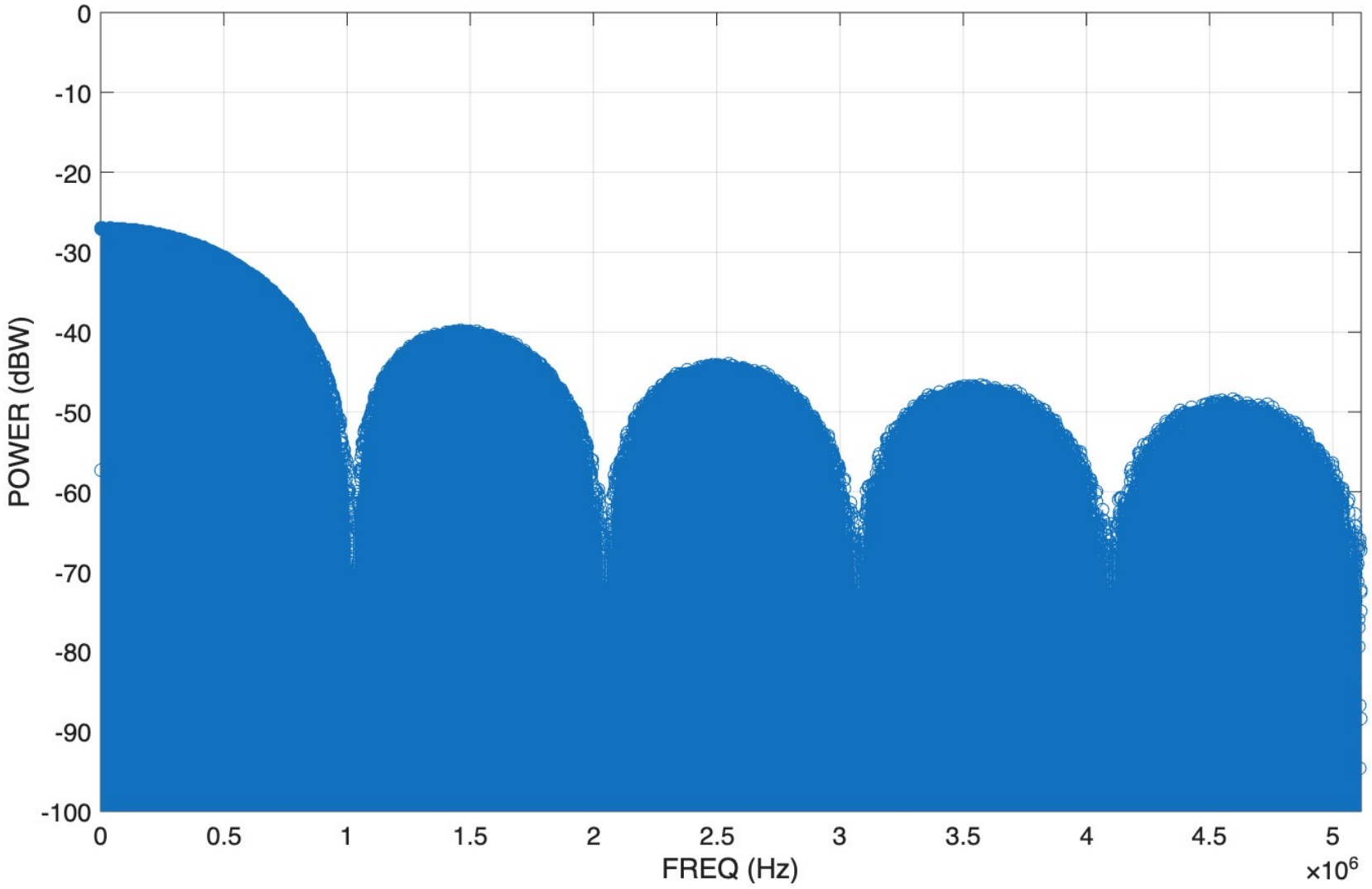
[~,x] = get_code([10 3]); codelength = length(x);
frep = fc/codelength; % This is the frequency(Hz) at which the code is periodic

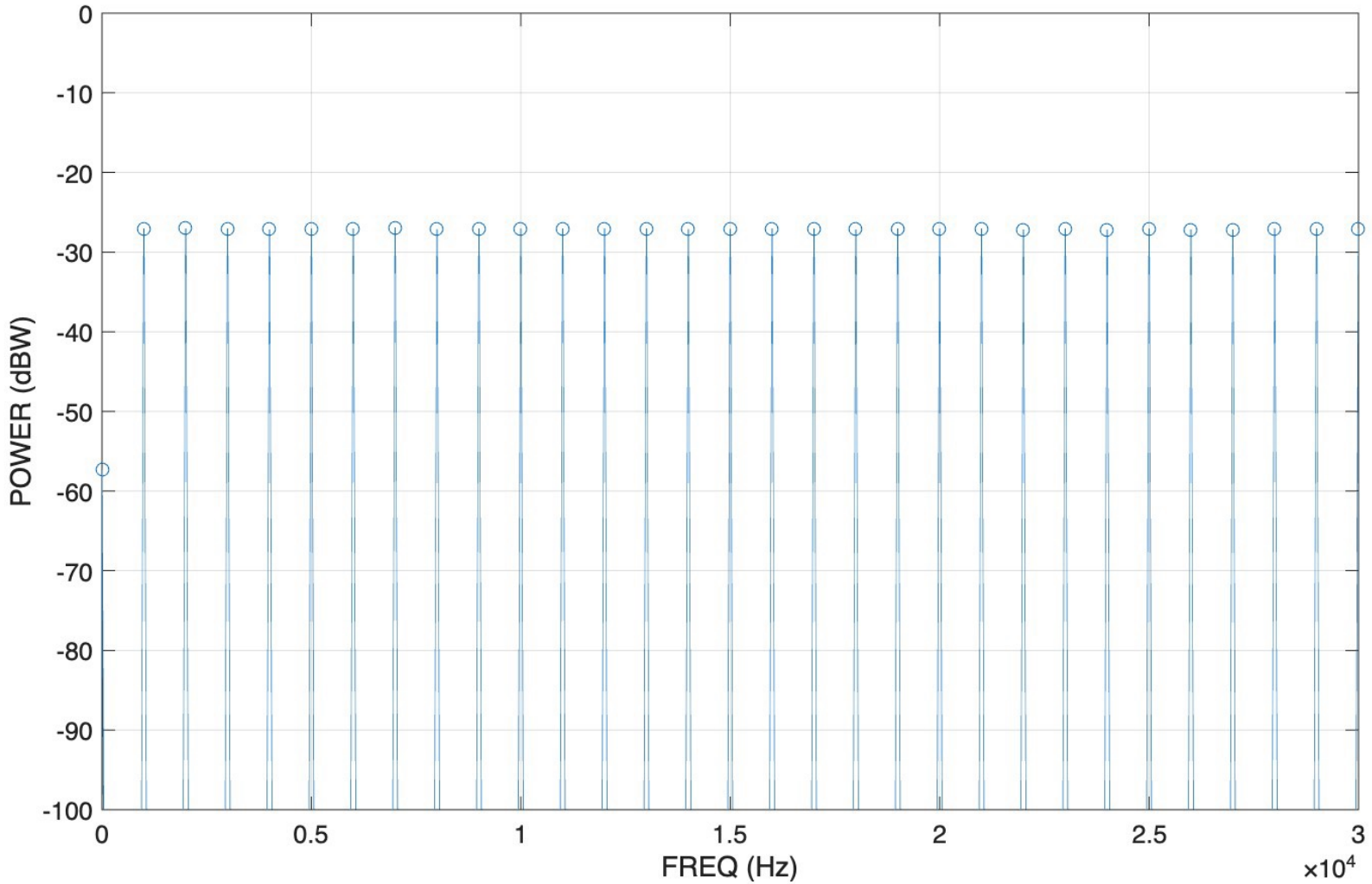
y4 = x(mod(floor(t/Tc),codelength)+1)';  %This gets the right code bit for each ts
f4 = fs*(0:npts-1)/npts;
py4 = 10*log10(2*abs(fft(y4)/npts).^2);
% Better way to compute it is: py = 20*log10(sqrt(2)*abs(fft(y4)/npts));

py4(py4<-100)=-300; % Get rid of -Inf so we can plot and see the spikes

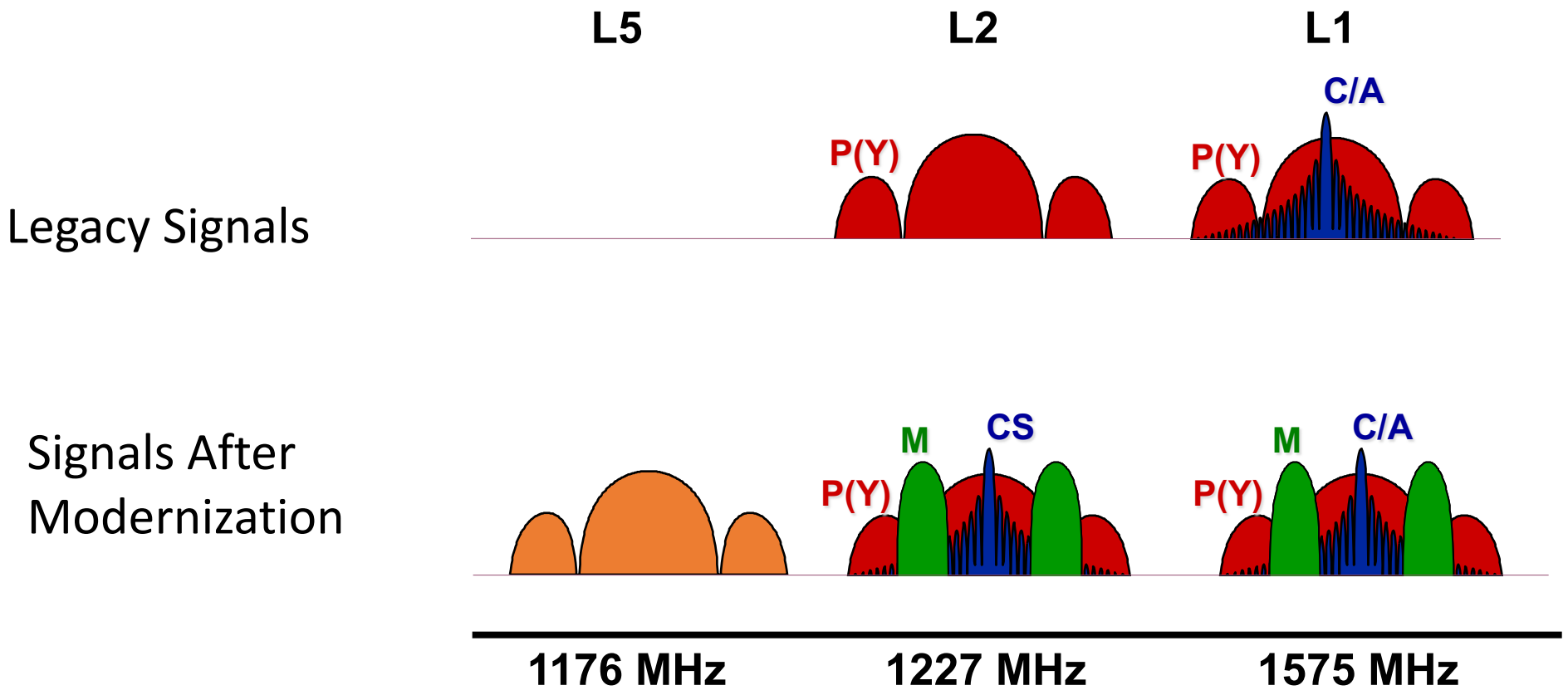
plot(f4,py4,'o-'),grid on, xlabel('FREQ (Hz)'), ylabel('POWER (dBW)')
xlim([0 fs/2]),ylim([-100 0])
```


0
-10
-20
-30
-40
-50
-60
-70
-80
-90
-100
POWER (dBW)
0
0.5
1
1.5
2
2.5
FREQ (Hz)
×10^7


0
-10
-20
-30
-40
-50
-60
-70
-80
-90
-100
POWER (dBW)
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
FREQ (Hz)
×10^6


0
-10
-20
-30
-40
-50
-60
-70
-80
-90
-100
POWER (dBW)
0
0.5
1
1.5
2
2.5
3
FREQ (Hz)
×10^4

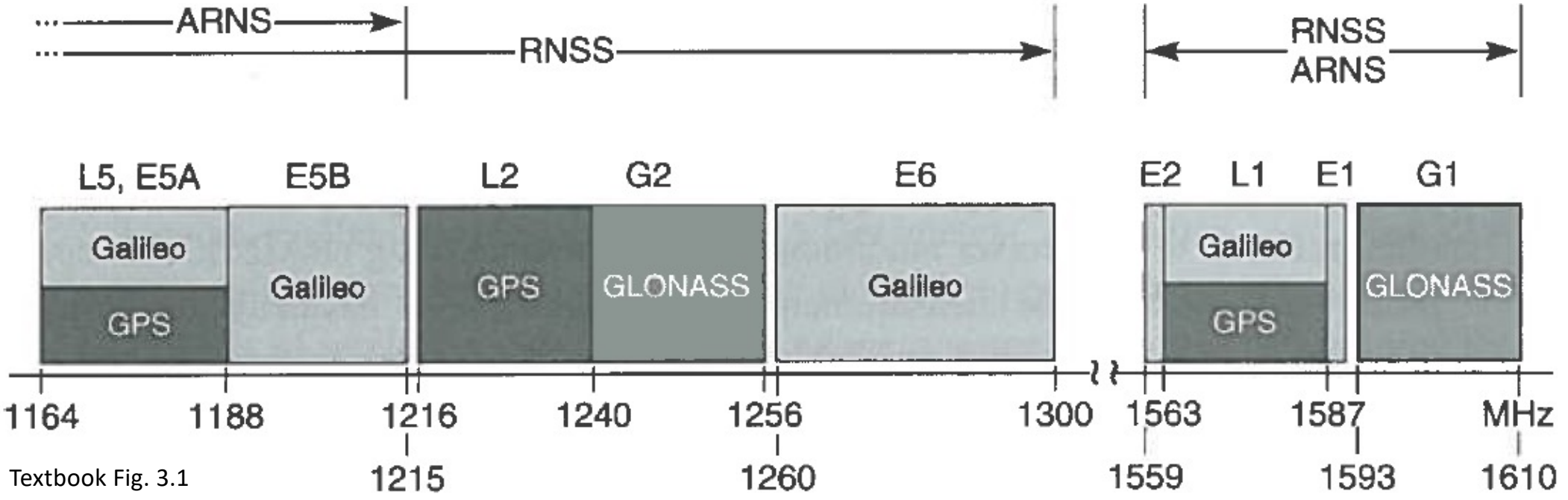
Modernized GPS Signal Evolution



From A.J. van Dierendonck

Frequency Allocations

- International Telecommunication Union (ITU), a United Nations organization, allocates frequencies for government and private sectors global telecom networks and services.
- Two bands of interests for navigation systems:
  - RNSS: Radio Navigation Satellite Services
  - ARNS: Aeronautical Radio Navigation Services -> more restricted, less interference, suitable for safety-for-life applications



Textbook Fig. 3.1

Spreading Codes and Ranging Signals Design Consideration

- Orthogonal signals
  - TDMA, FDMA, CDMA, …
- Ranging signal components:
  - RF carrier: frequency band allocation
  - Navigation data message: higher bit rate → requires higher SNR
  - Spreading code:
    - **Rate**: Faster code → wider bandwidth
      Pros: sharper correlation peak + less multipath impact→ better precise time estimation
      Cons: higher sampling rate + higher computation load + higher power consumption
    - **Code**: Longer code → lower cross-correlation
      Pros: less interference among satellite signals
      Cons: more computation requirements
    - **Pulse shape**: rectangular pulse with +1 and -1 amplitude
    - **Modulation**: see next slide for details

Features of Modernized Signals

- Two and three center frequencies
- Pilot components not modulated by data
- Longer spreading codes; overlay codes on pilot components
- Higher transmit power
- Strong error correction; very strong error detection
- Wider bandwidth spreading modulations using BOC
- Highly interoperable signals to be transmitted by multiple systems—up to 100 satellites

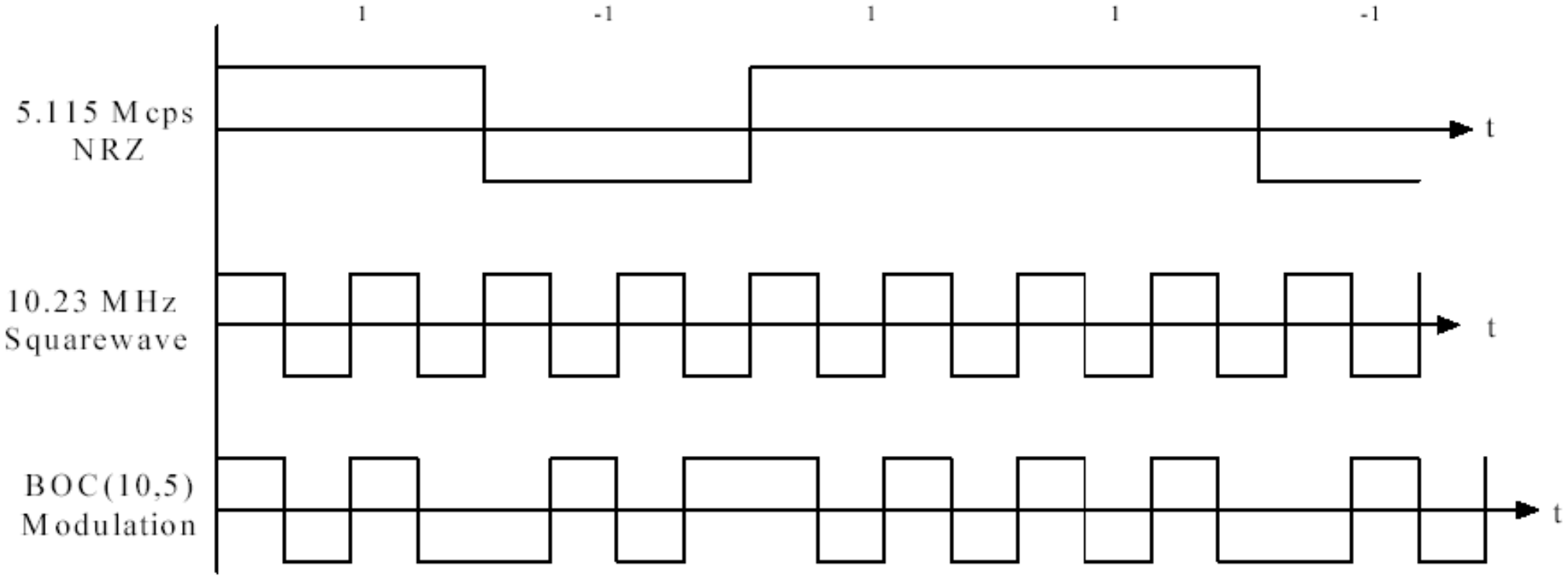
FROM BETZ presentation 17 March 2011 (pnt.gov)

Binary Offset Carrier (BOC) Signals

- BOC(α, β) Notation
    - f0 = 10,230,000 Hz
    - f1 = f0/10 = 1,023,000 Hz
    - fs = fsquare = α * f1
    - fc = fch = β * f1
- The new military M-code is BOC(10,5)
    - fsquare = 10 * f1 = 10,230,000 Hz
    - fch = 5 * f1 = 5,115,000 Hz
- The new L1C signals use BOC(1,1) and others….
- BOC(m,m) is also called a Manchester Code

BOC signals have different spectra and autocorrelations than maximal length or Gold Codes

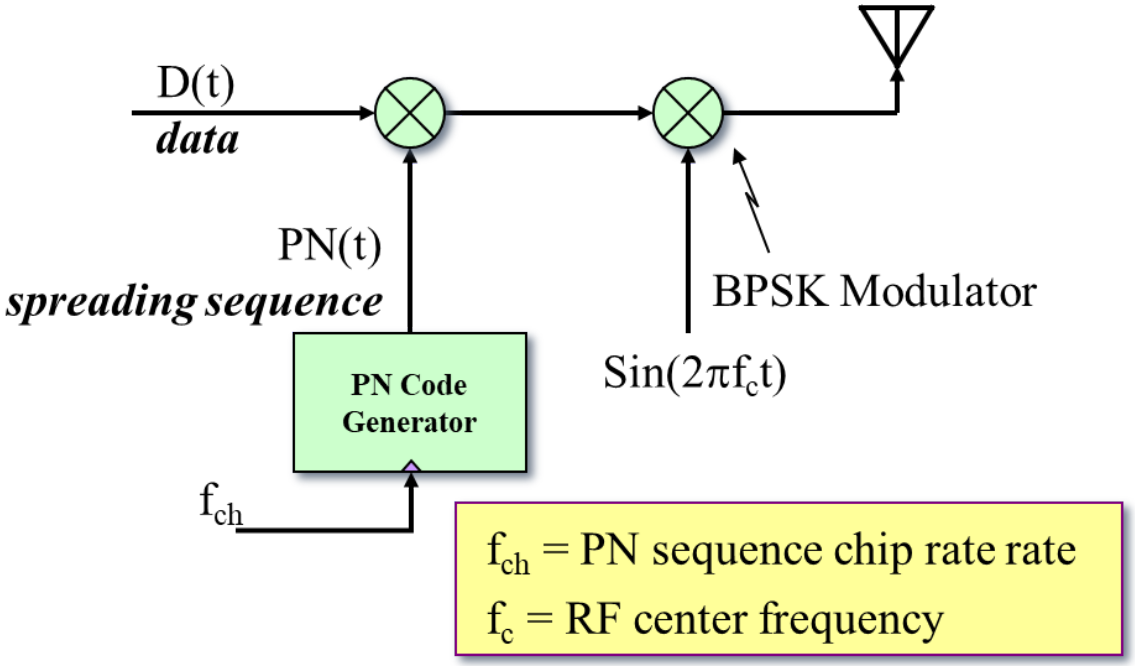
BOC(10,5) Modulation and Its Components



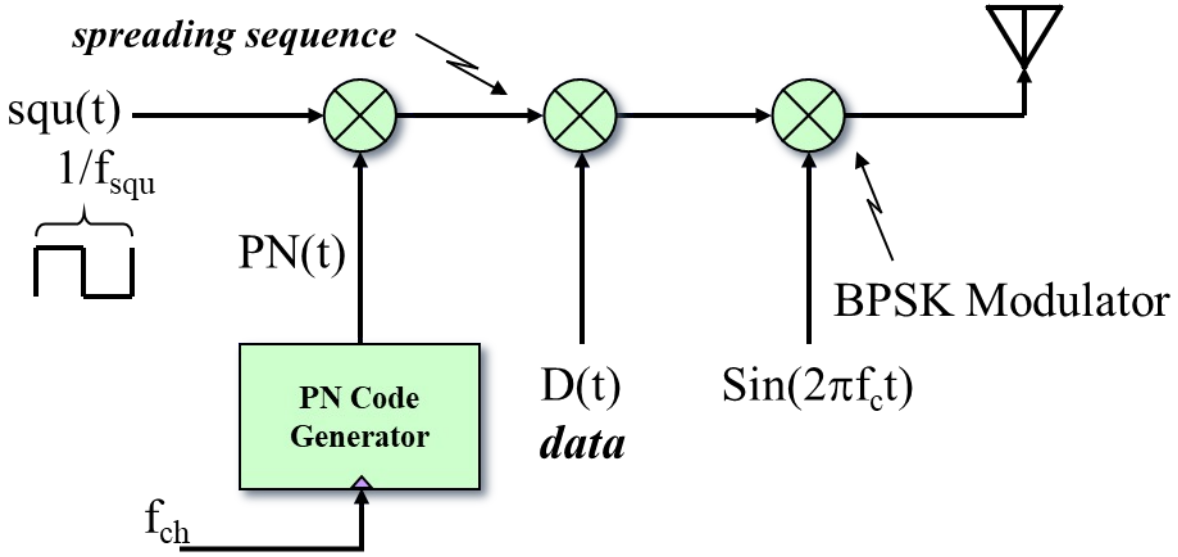
From: Holmes and Dafish "Matched Filter Mean Acquisition Time Performance For P(Y), BOC(10,5), and MAN(10) Codes With FFT Aiding and Noncoherent Combining" ION 57th Annual Meeting/CIGTF 20th Biennial Guidance Test Symposium, 11-13 June 2001, Albuquerque, NM

Simplified BPSK and BOC Signal Generation

- **BPSK**



- **BOC**



f_{squ} = subcarrier frequency = square wave freq

f_{ch} = PN sequence chip rate rate

f_c = RF center frequency

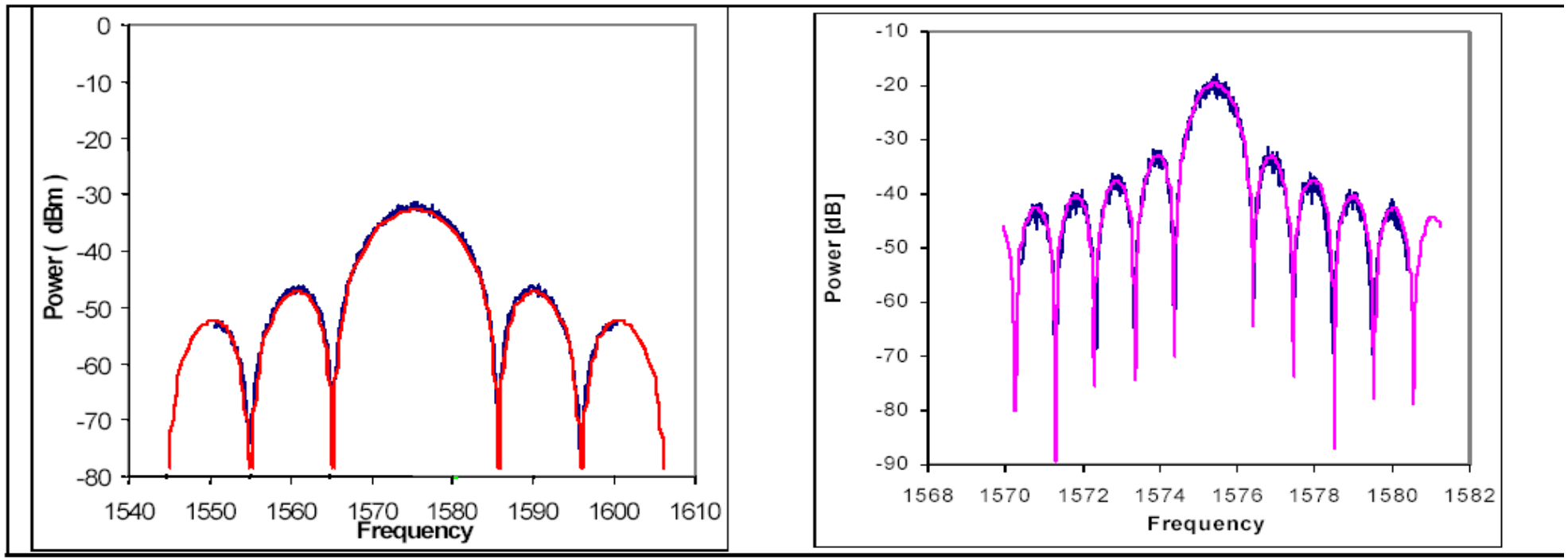
Courtesy Logan Scott

PRN Code Modulated Signal - Reminder

- Dominant feature is the sinc function

$$F\{x(t)\} = T_c \ \mathrm{sinc}\left(\pi\left(f - f_{L1}\right)T_C\right)X\left(f - f_{L1}\right)$$

$$= T_c \ \mathrm{sinc}\left(\frac{\pi\left(f - f_{L1}\right)}{f_c}\right)X\left(f - f_{L1}\right)$$



SPECTRUM of BOC(α, β)

- fs = square wave frequency (αf_0)
- fc = PRN code chip rate (βf_0)

$$S_{BOC_S}(f) = T_c \frac{\sin^2(\pi f / f_c)}{(\pi f / f_c)^2} \tan^2\left(\frac{1}{2}\pi f / f_s\right)$$

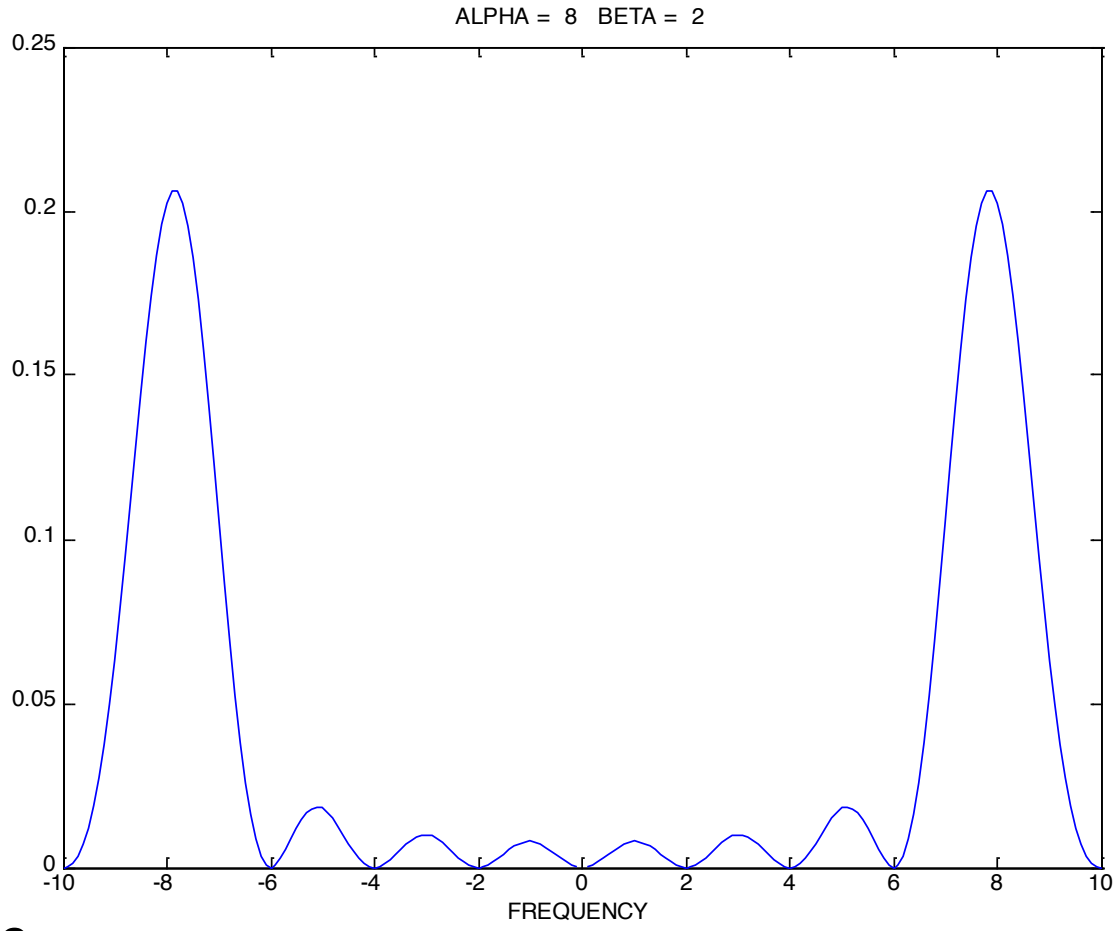
To find the width of the lobes, determine where this goes to zero.

From Kaplan and Hegarty, p. 121

BOC Examples: α=8, β=2

Peak is offset from center by fsquare

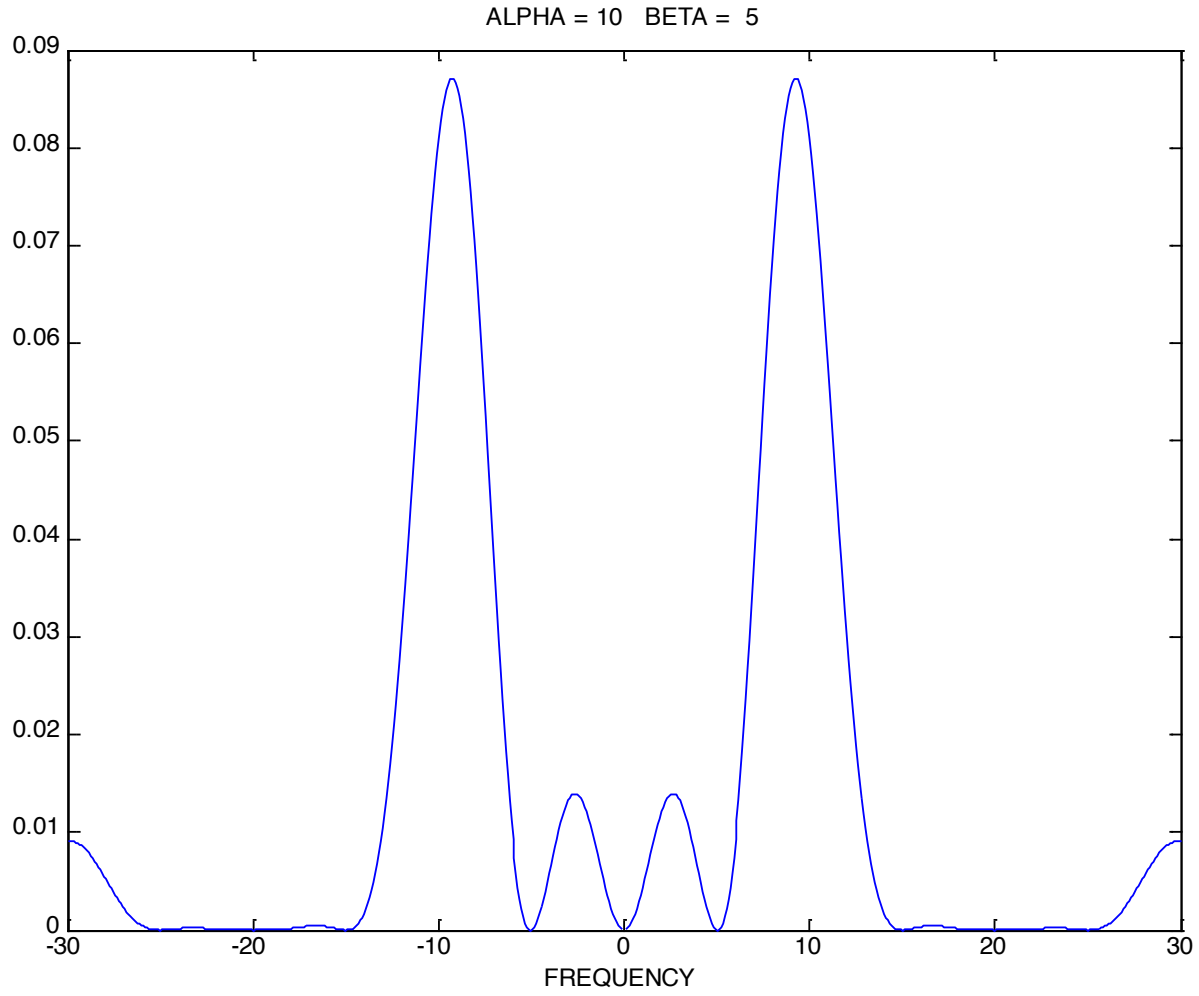
Side lobe width = fcode

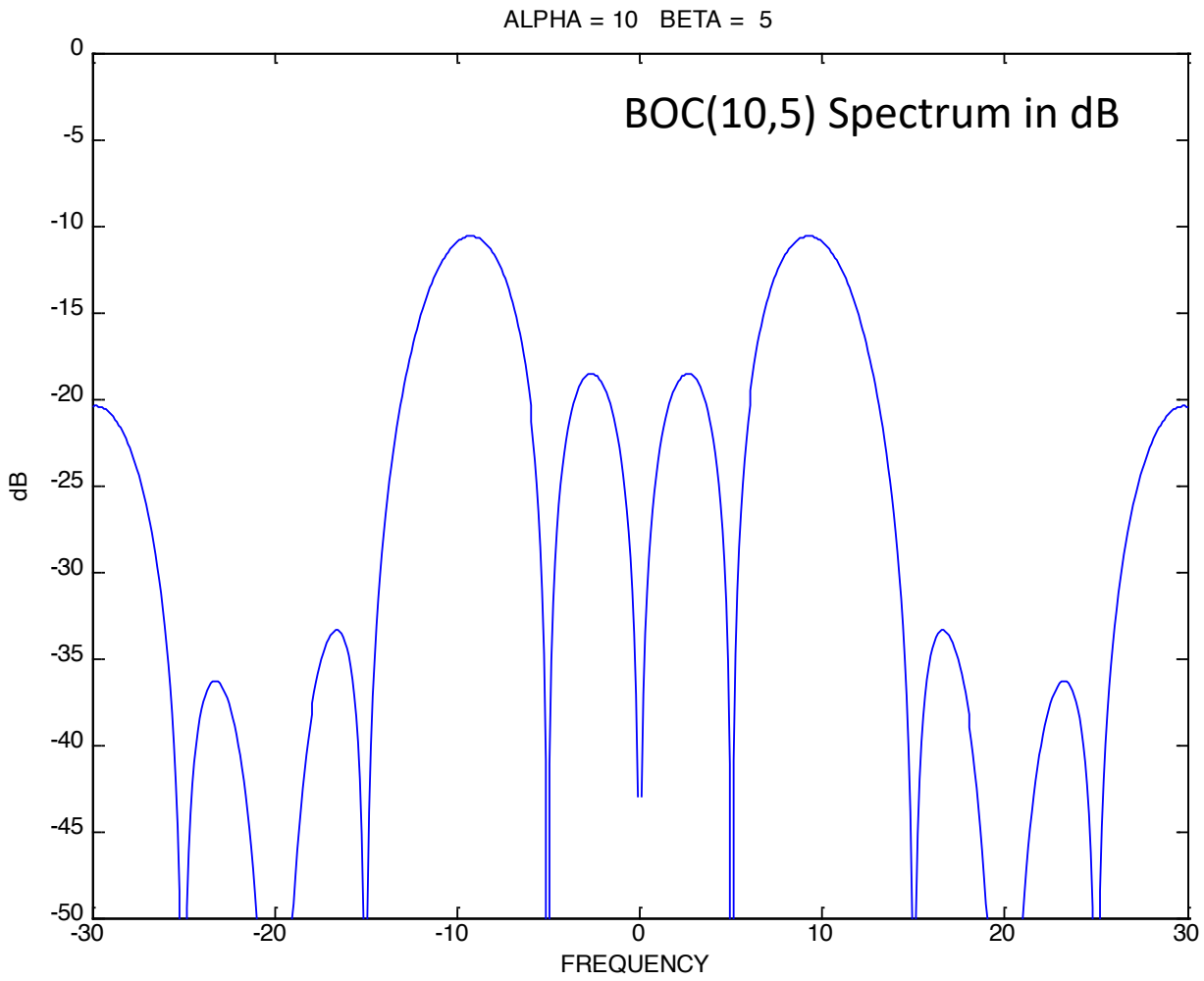


Main lobe width is usually 2xfcode, but depends on the tan term shown on previous slide

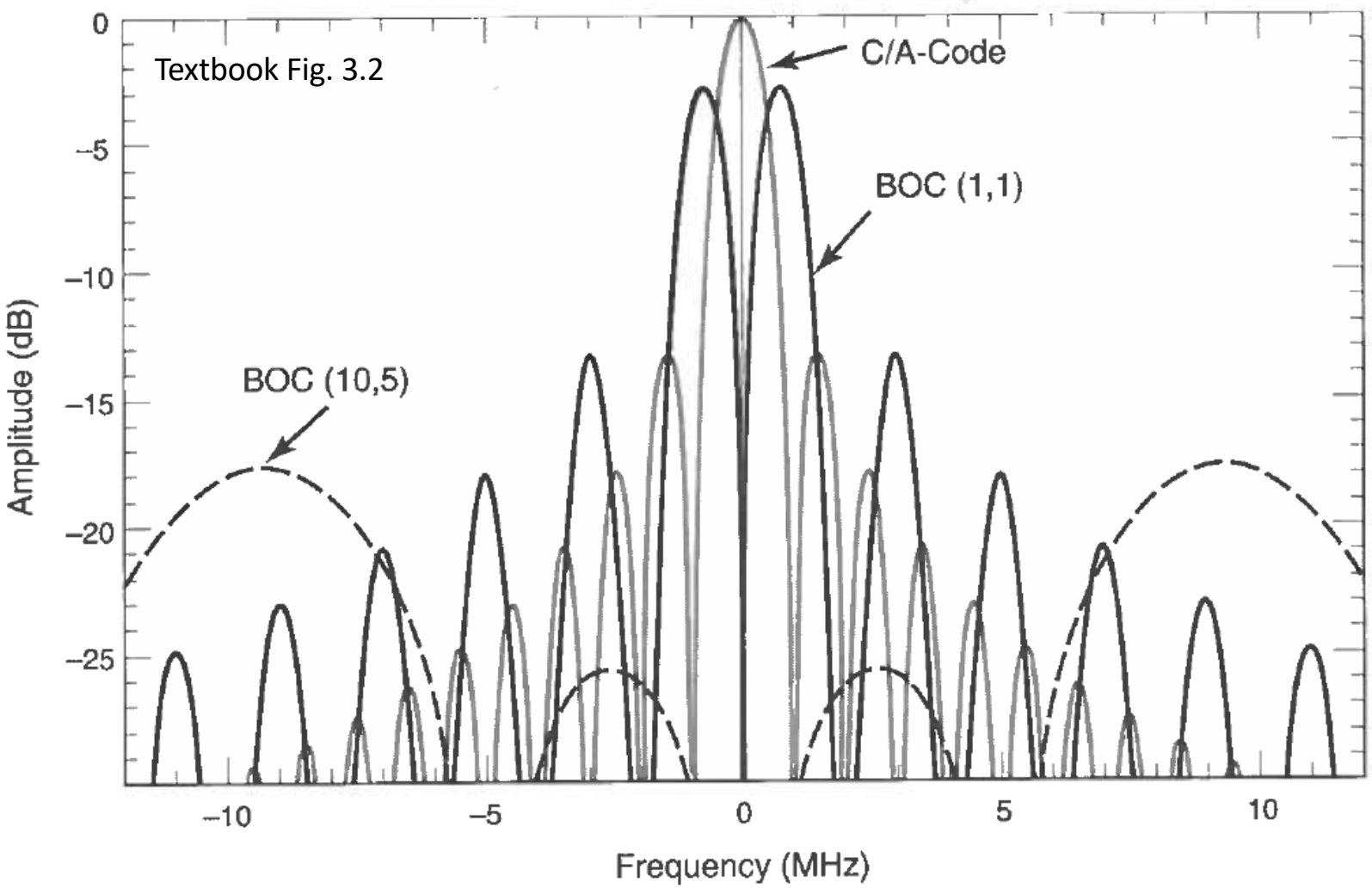
When fcode & fsquare are the same,
the main lobe is split into two parts – each has width of 2*fcode.

BOC Examples: α=10, β=5





C/A Code vs. BOC (binary offset carrier)



R-n code: Rectangular pulse code, *n* = multiples of 1.023MHz

BOC(*m*, *n*): binary offset carrier, the product of an *R-n* code multiplied by a subcarrier of frequency *m*x1.023MHz

Autocorrelation comparison C/A and BOC

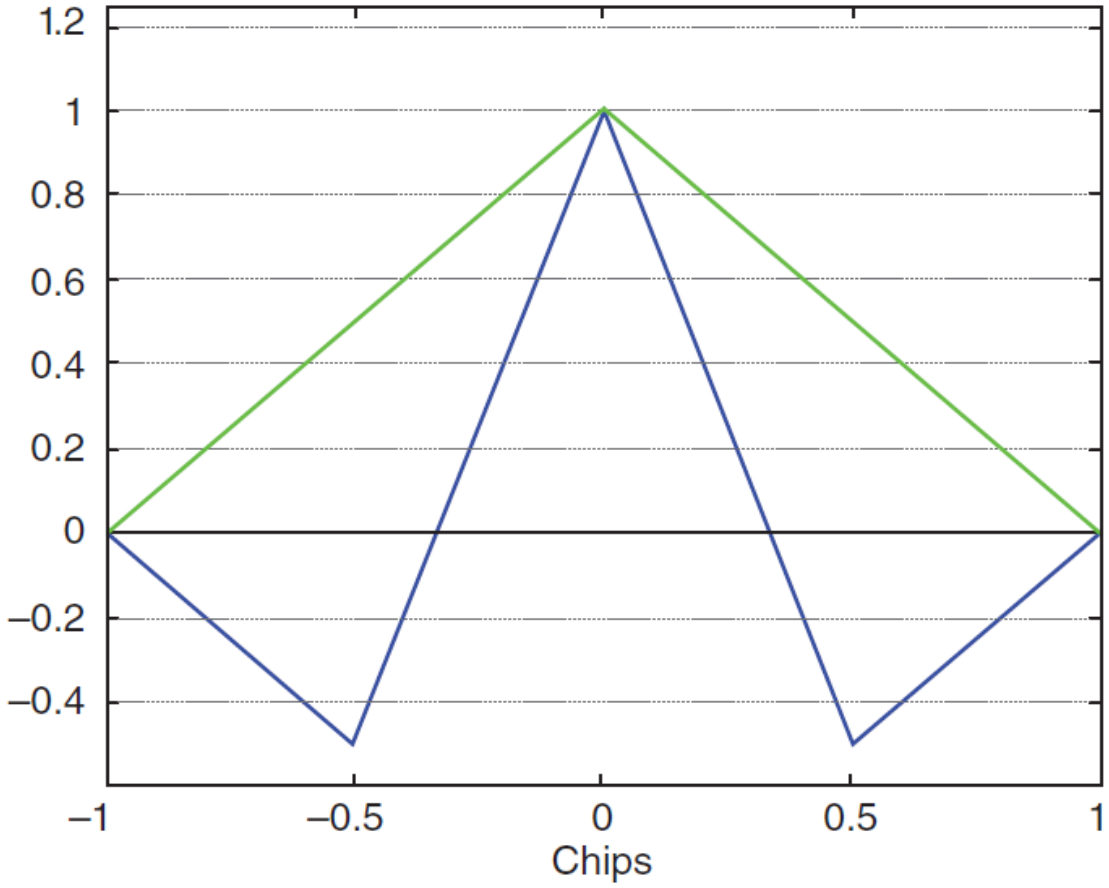


Figure 9.6 Autocorrelation function of BPSK(1) and BOC(1,1) signals.

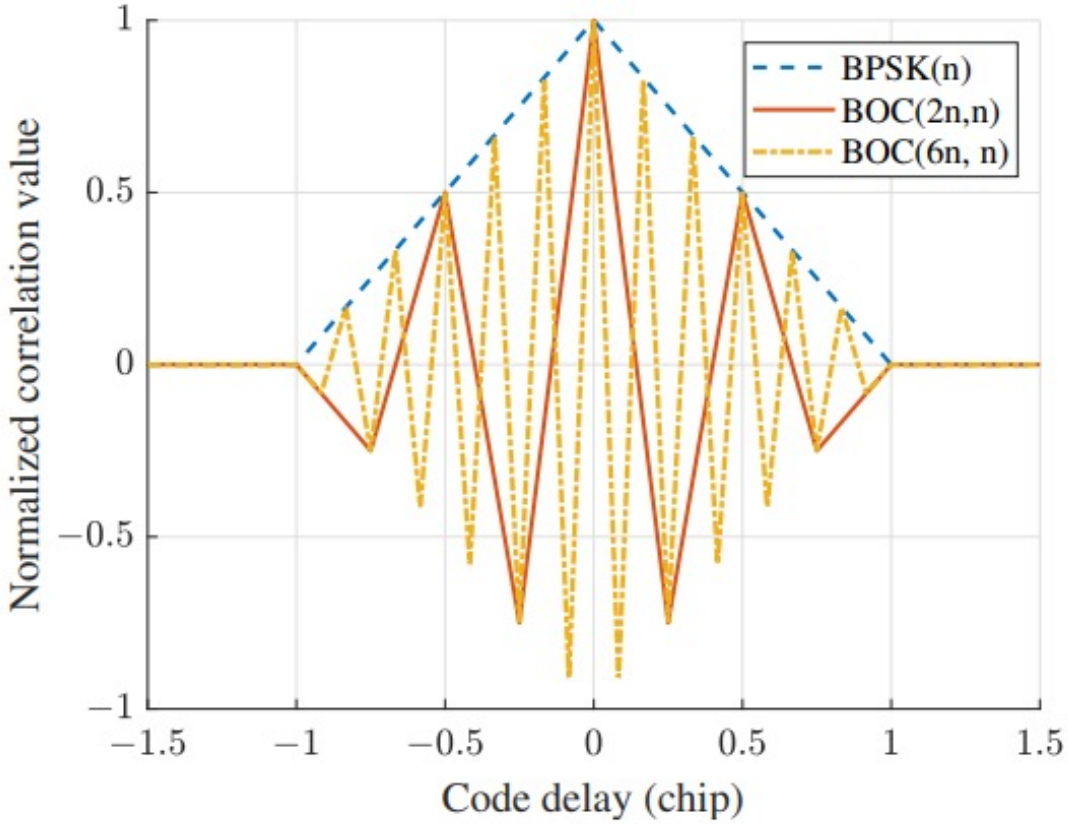


Figure 1. Normalized auto-correlation functions of BPSK(n), BOC(2n,n), and

(Tian et al., 2023)

GPS Modernization

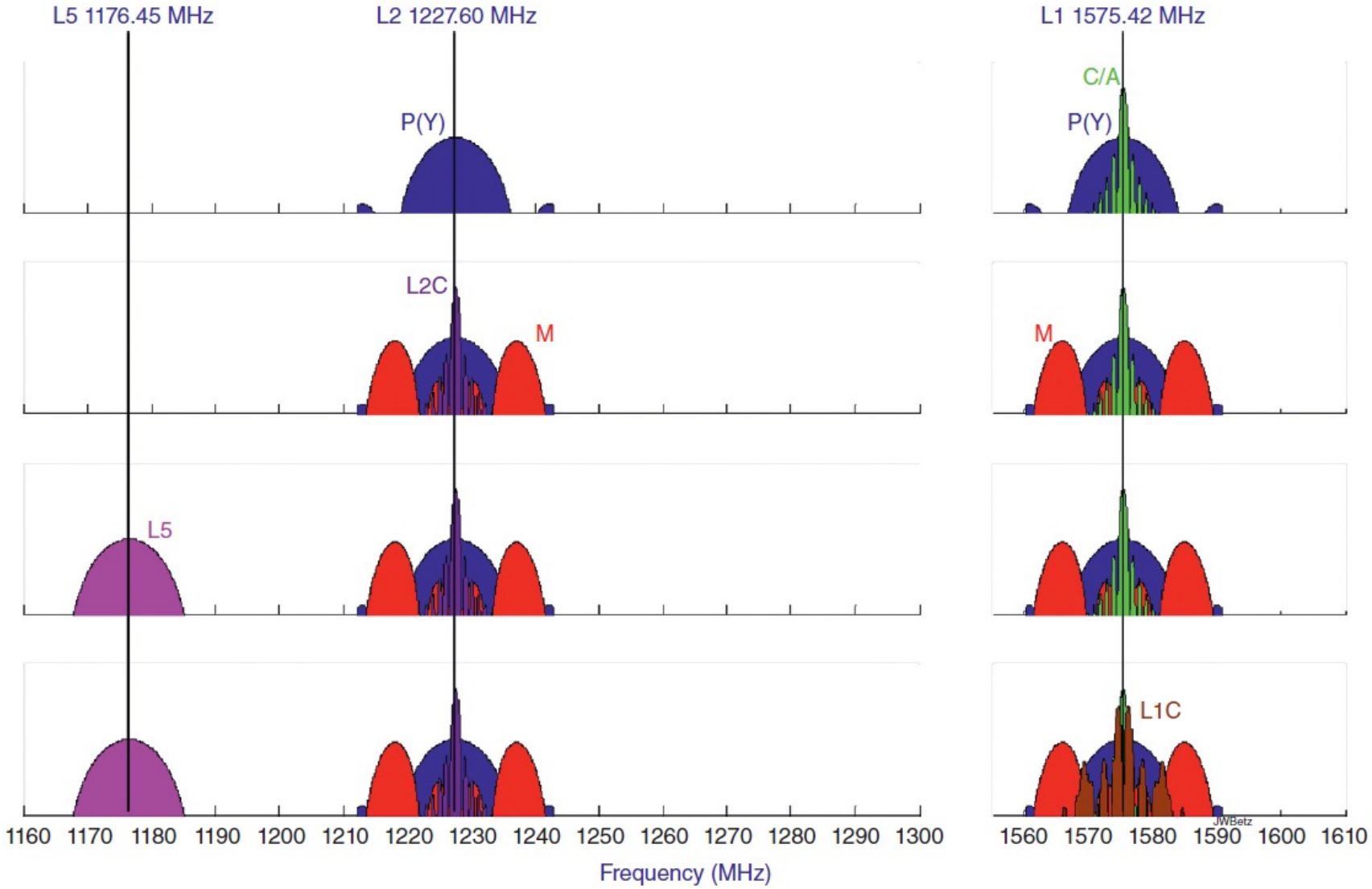


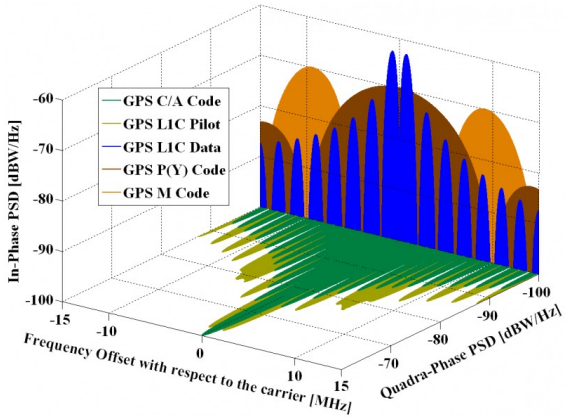
Figure 3.1 Modernization of GPS signals [25]. Source: Reproduced with permission of John Wiley & Sons.

Satellites
Block II / IIA / IIR 1989-2005
Block IIR-M Starting in 2005
Block IIF Starting in 2009
Block III Starting in 2018

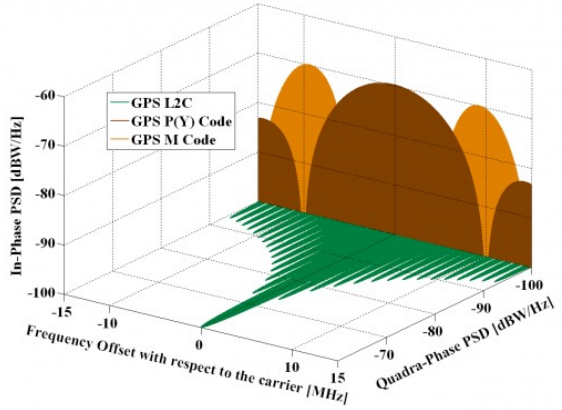
Modernized GPS signals

- Newest block of GPS satellites (GPS-III)
  - 4 civil signals: L1 C/A, **L1C**, L2C, L5 •
    - First GPS SVs with interoperable L1C signal
  - 4 military signals: L1/L2 P(Y), L1/L2M
  - **3 improved Rubidium atomic clocks**

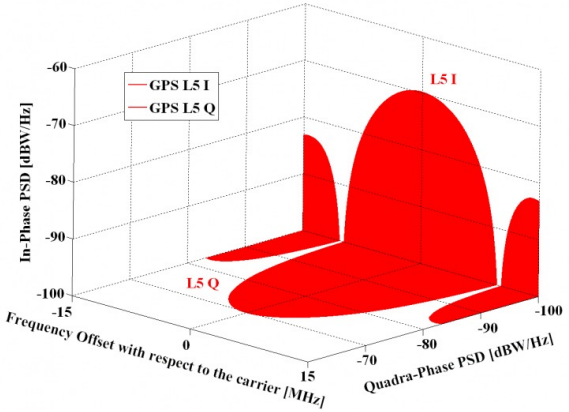
L1 Frequency



L2 Frequency



L5 Frequency



Images source: www.navipedia.net

L2C Civil Signal



- Center Frequency of 1227.6 MHz
- Improves PVT for scientific/commercial dual frequency users
  - Can measure ionospheric delay
- Superior code cross correlation and continuous wave (CW) interference rejection (relative to C/A's 24 dB)
- Improved data structure for enhanced data demodulation (5 dB better than C/A)
- Data Free Pilot Channel for Improved Tracking
- Enhanced multipath rejection using joint L1/L2/L5 signal processing
- Better indoor performance is possible through channel diversity gains
- Not supported by WAAS (spectrum issue)

From Logan Scott

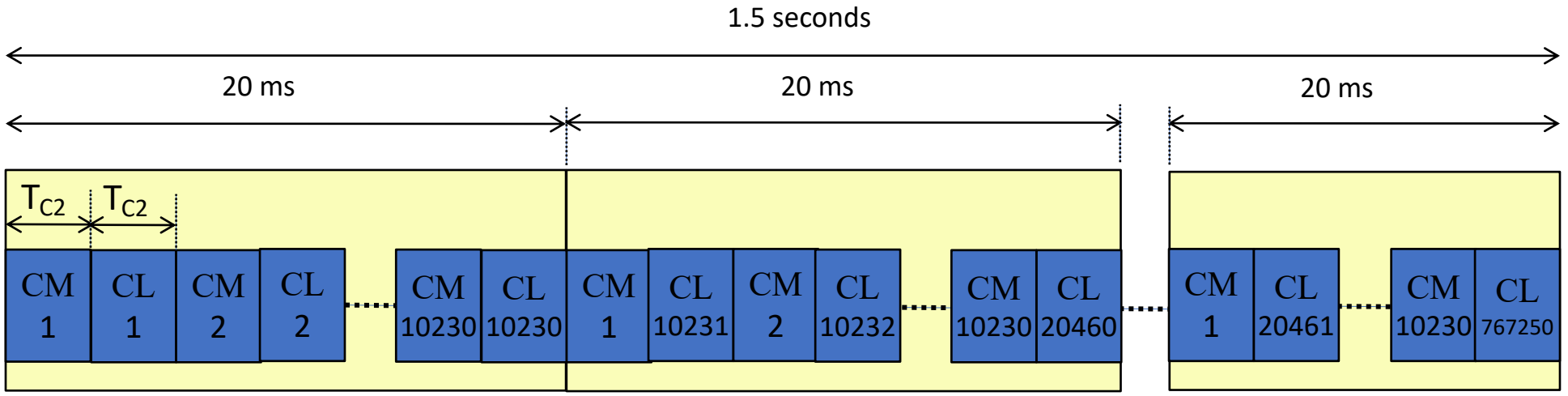
L2C Signal Details (1)

- R-1 code (1.023MHz chipping rate)
- Two PRN codes multiplexed chip by chip:
  - CM → moderate length code with 511.5kHz chipping rate
  - CL → long code with 511.5kHz chipping rate
  - Clock rate: 1.023MHz
- Code length:
  - CM: 10,230 chips → what is the length in time?
  - CL: 767,250 chips → what is the length in time?
- Navigation data:
  - CM: 50 symbols per second with ½ forward error correction (FEC) code → 25 Hz effective bps rate

L2C Signal Details (2)

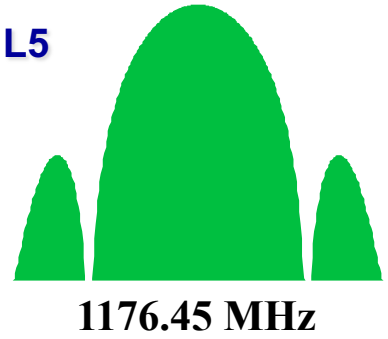
$$s_{L2C}^{(k)}(t) = \sqrt{2P_{C2}}\; D_{C2}^{(k)}(t) CM^{(k)}(t) \cos(2\pi f_{L2} t + \theta_{L2}), \quad nT_{C2} < t \le (n+1/2)T_{C2}$$

$$s_{L2C}^{(k)}(t) = \sqrt{2P_{C2}}\; CL^{(k)}(t) \cos(2\pi f_{L2} t + \theta_{L2}), \quad (n+1/2)T_{C2} < t \le (n+1)T_{C2}$$



T_{C2} = 1/1.023MHz ~997 ns

L5 Civil Signal



- Center Frequency 1176.45 MHz
- Improves signal structure for enhanced performance
  - Higher power -154 dBW (not implemented in Block IIF)
  - Wider bandwidth 10.23 MCh/sec
  - Longer spreading codes in the navigation message
  - Use of Neumann-Hoffmann codes to reduce spectral line spacing
- Aeronautical Radionavigation Services band
- L5 signal definition in IS-GPS-705
- Plans to provide L5 GEO Signals as part of WAAS

L5 Signal Details

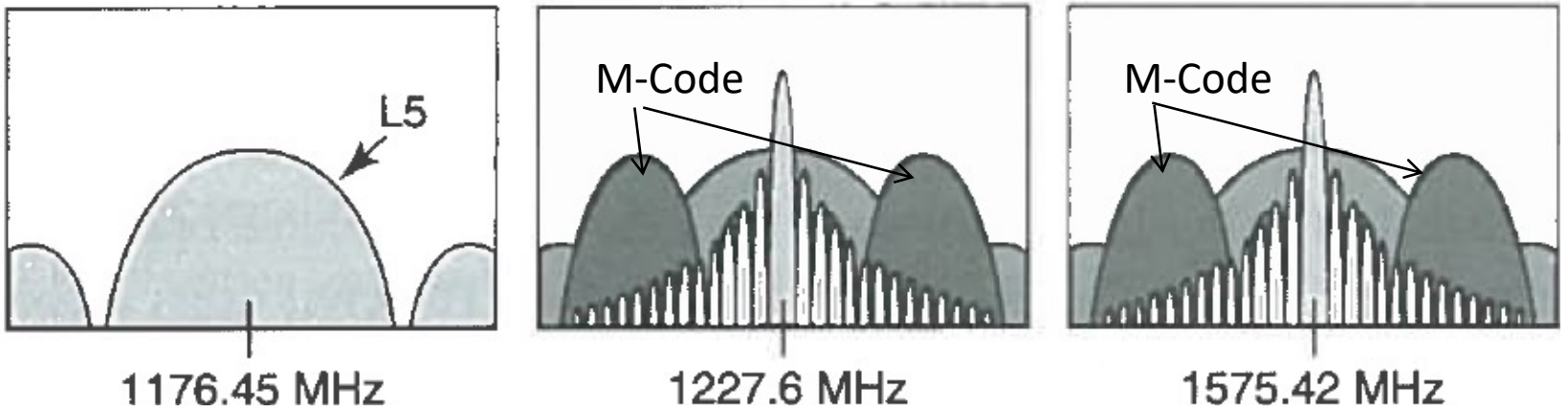
$$s_{L5}^{(k)}(t) = \sqrt{2P_{L5}}\, D_{L5}^{(k)}(t)\, h_I(t)\, g_I^{(k)}(t) \cos\left(2\pi f_{L5} t + \theta_{L5}\right)$$
$$+\sqrt{2P_{L5}}\, h_Q(t)\, g_Q^{(k)}(t) \sin\left(2\pi f_{L5} t + \theta_{L5}\right)$$

No nav data on Q5

1. Clock rate: 10.23MHz
2. Code length: 10230 chips → 1ms
3. Product code: both codes are multiplied by overlay codes (Neuman-Hofman): a sequence of 1ms rectangular pulses with +1 and -1 amplitude
   - g_I is multiplied with a 10 ms long sequence h_I
   - g_Q is multipled with a 20 ms long sequence h_Q

New Military Signals (M-Codes)

- Have to co-exist with civilian signals in the same bands, 24MHz bandwidth
- Have sufficient spectrum isolation to prevent mutual interference and to allow civil signal jamming.
- Better jamming resistance than P(Y) code signals ← higher power
- Additional high gain antenna to “spot beam” at a specific region
- More robust acquisition ← direct acquisition possible
- New navigation data message (MNAV) ← message-based com protocol, more flexible.
- New architecture for SV signal generation and transmission, no correlation between L1 and L2 signals.
- Power: -158dBW for whole earth, -138 dBW for spot beam



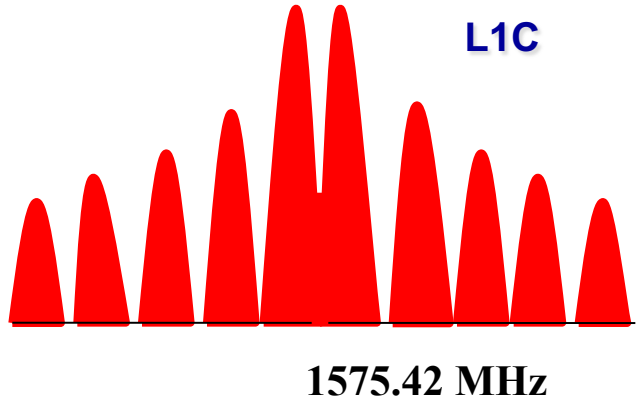
GPS III

- Higher signal power with “spot beam” antennas
- Laser retro-reflectors to track SV orbits independent of radio signals, thereby separating clock error from ephemeris error – starting w/SV09.
- Include an additional civilian signal on L1: L1C
- Accompanied with modernized control segment
- L1C: higher power (1.5 dB increase compared to C/A)
    - Located at L1 band (common for current GPS L1, Galileo, QZSS, SBAS)
    - 75% power in pilot signal
    - Weil-based spreading codes
    - An overlay code on pilot signal for data message synchronization
    - Advanced error correction for nav data
    - Data symbol interleaving to combat fading
- More readings about L1C: http://www.insidegnss.com/node/485, IS-GPS-800

L1C Civil Signal Compatible with Galileo & QZSS

- Adds a modernized L1 civil signal with data free Pilot
- Continue to provide C/A code to ensure backward compatibility
- Greater civil interoperability with Galileo
- Block III Capability (1st Launch 2016)
- On L1C, the power split is 3/4 on the Pilot and 1/4 on the data channel, for more robust tracking in challenged environments
- MBOC - BOC(1,1) & BOC(6,1) – different implementations

The MBOC Modulation, InsideGNSS, Sept/Oct 2007
http://www.insidegnss.com/node/174



5

GPS Modernization

SPACE AND MISSILE SYSTEMS CENTER

Space Segment

SV families provide L-Band broadcast to User Segment

GPS IIA/IIR
- Basic GPS
- Nuclear Detonation Detection System (NDS)

GPS IIR-M
- 2nd Civil Signal (L2C)
- New Military Signal
- Increased Anti-Jam Power

GPS IIF
- 3rd Civil Signal (L5)
- Longer Life
- Better Clocks

GPS III (SV01-10)
- Accuracy & Power
- Increased Anti-Jam Power
- Inherent Signal Integrity
- 4th Civil Signal (L1C)
- Longer Life
- Better Clocks

GPS IIIF (SV11-32)
- Unified S-Band Telemetry, Tracking & Commanding
- Search & Rescue (SAR) Payload
- Laser Retroreflector Array
- Redesigned NDS Payload

Control Segment

TT&C of Space Segment assets & distribution of data to user interfaces

Legacy (OCS)
- Mainframe System
- Command & Control
- Signal Monitoring

Architecture Evolution Plan (AEP)
- Distributed Architecture
- Increased Signal Monitoring Coverage
- Security
- Accuracy

OCX Block 0
- GPS III Launch & Checkout System

GPS III Contingency Ops (COps)
- GPS III Mission on AEP

M-Code Early Use (MCEU)
- Update OCS to operationalize Core M-Code

OCX Block 1/2
- Fly Constellation & GPS III
- Begin New Signal Control
- Upgraded Information Assurance

OCX Block 2+
- Control all signals
- Capability On-Ramps
- GPS IIIF Evolution

User Segment

Applies Space and Control Segment data for PNT applications

Continued support to an ever-growing number of applications
- Annual Public Interface Control Working Group (ICWG)
- Standard Positioning Service (SPS) Performance Standard Updates
- Precise Positioning Service (PPS) Enhancements
- Sustained commitment to transparency
- Visit GPS.gov for more info

Modernized Civil Signals
- L2C (Various commercial applications)
- L5 (Safety-of-life, frequency band protected)
- L1C (Multi-GNSS interoperability)

Source: https://www.gps.gov/cgsic/meetings/2020/colburn.pdf

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Next lecture: Receiver Overview, Antennas & Link Budget